

Exercise J1

Q1. If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on x and y axis respectively, then the value of a^2b is

- (A) $27c^3$ (B) $\frac{4}{27}c^3$ (C) $\frac{27}{4}c^3$ (D) $\frac{4}{9}c^3$

Ans. C

Sol. $x^2y = c^3$

$$x^2 \frac{dy}{dx} + 2xy = 0 \Rightarrow \frac{dy}{dx} = -\frac{2y}{x}$$

equation of tangent at (x, y)

$$Y - y = -\frac{2y}{x}(X - x)$$

$$Y = 0, \text{ gives } X = \frac{3x}{2} = a$$

and $X = 0$, gives, $Y = 3y = b$

$$\text{Now } a^2b = \frac{9x^2}{4} \cdot 3y = \frac{27}{x^2y} = \frac{27}{4}c^3$$

Q2. At the point $P(a, a^n)$ on the graph of $y = x^n$ ($n \in \mathbb{N}$) in the first quadrant a normal is drawn. The normal intersects the y -axis at the point $(0, b)$. If $\lim_{a \rightarrow 0} b = \frac{1}{2}$, then n equals

- (A) 1 (B) 3 (C) 2 (D) 4

Ans. C

Sol. -

Q3. The normal to the curve $x = a(1 + \cos \theta)$, $y = a \sin \theta$ at ' θ ' always passes through the fixed point-

- (A) $(a, 0)$ (B) $(0, a)$ (C) $(0, 0)$ (D) (a, a)

Ans. A

Sol. $x = a(1 + \cos \theta) = a \sin \theta$

$$\frac{dx}{d\theta} = -a \sin \theta \quad ; \quad \frac{dy}{d\theta} = a \cos \theta$$

$$\left(\frac{dy}{dx} \right) = \frac{\cos \theta}{\sin \theta}$$

$$\text{slope of normal} = -\left(\frac{dx}{dy} \right) = \frac{\sin \theta}{\cos \theta}$$

$$y - a \sin \theta = \frac{\sin \theta}{\cos \theta}(x - a - a \cos \theta)$$

$$y\cos\theta - a\sin\theta\cos\theta = x(\sin\theta)(1 + \cos\theta)$$

$$x\sin\theta - y\cos\theta = a\sin\theta(1 + \cos\theta - \cos\theta)$$

clearly passes through $(a, 0)$

Q4. The normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at $(1, 1)$:

- (A) does not meet the curve again (B) meets the curve again in the second quadrant
(C) meets the curve again in the third quadrant
(D) meets the curve again in the fourth quadrant

Ans. D

Sol. $(x + 3y)(x - y) = 0$

Equation of normal at $(1, 1)$ $x + y = 2$

will meet the curve again at $(3, -1)$

Q5. Consider $f(x) = \tan^{-1} \left(\sqrt{\frac{1 + \sin x}{1 - \sin x}} \right)$, $x \in (0, \frac{\pi}{2})$. A normal to $y = f(x)$ at $x = \frac{\pi}{6}$ also passes through the point :

- (A) $(\frac{\pi}{6}, 0)$ (B) $(\frac{\pi}{4}, 0)$ (C) $(0, 0)$ (D) $(0, \frac{2\pi}{3})$

Ans. D

Sol. $f(x) = \tan^{-1} \left(\frac{1 + \sin x}{\cos x} \right)$

$$= \tan^{-1} \left(\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right)$$

$$f'(x) = \frac{1}{2}$$

Eq. of normal $y - \frac{\pi}{3} = -2 \left(x - \frac{\pi}{6} \right)$

$$2x + y = \frac{2\pi}{3}$$

Q6. The normal to the curve $y(x - 2)(x - 3) = x + 6$ at the point where the curve intersects the y-axis passes through the point

- (A) $(\frac{1}{2}, -\frac{1}{3})$ (B) $(\frac{1}{2}, \frac{1}{3})$ (C) $(-\frac{1}{2}, -\frac{1}{2})$ (D) $(\frac{1}{2}, \frac{1}{2})$

Ans. D

Sol. Point $(0, 1)$

$$y(x - 2)(x - 3) = x + 6$$

$$\left(\frac{dy}{dx} \right) (x - 2)(x - 3) + (2x - 5)y = 1$$

Slope of tangent at $(0, 1) = 1$

Equation of normal

$$y - 1 = -1(x - 0)$$

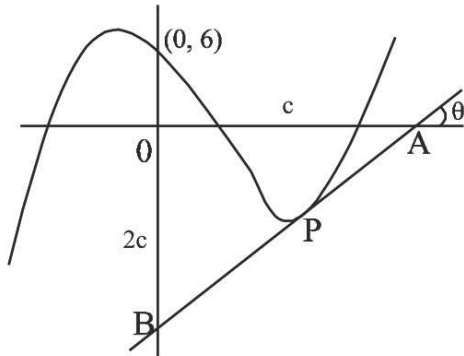
$$x + y = 1$$

Q7. The point(s) at each of which the tangents to the curve $y = x^3 - 3x^2 - 7x + 6$ cut off on the positive semi axis OX a line segment half that on the negative semi axis OY then the co-ordinates the point(s) is/are given by :

- (A) $(-1, 9)$ (B) $(3, -15)$ (C) $(1, -3)$ (D) none

Ans. B

Sol.



$$\text{If } OA = a ; OB = 2a \Rightarrow \tan\theta = 2$$

slope of the tangent is 2

$$\Rightarrow \left(\frac{dy}{dx}\right) = 2 \Rightarrow 3x_1^2 - 6x_1 - 7 = 2 \Rightarrow 3x_1^2 - 6x_1 - 9 = 0$$

$$\Rightarrow x_1 = 3 \text{ or } -1 \text{ (rejected)}$$

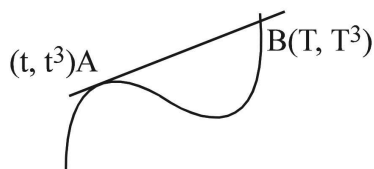
$$\Rightarrow (3, -15) \Rightarrow B$$

Q8. Let C be the curve $y = x^3$ (where x takes all real values). The tangent at A meets the curve again at B. If the gradient at B is K times the gradient at A then K is equal to

- (A) 4 (B) 2 (C) 2 (D) $\frac{1}{4}$

Ans. A

Sol.



$$\frac{dy}{dx} = 3x^2 = 3t^2 \text{ at 'A'}$$

$$\therefore 3t^2 = \frac{T^2 - t^3}{T - t} = T^2 + Tt + t^2$$

$$T^2 + Tt - 2t^2 = 0$$

$$(T - t)(T + 2t) = 0 \Rightarrow T = t \text{ or } T = -2t \text{ (} T = t \text{ is not possible)}$$

$$\frac{m_B}{m_A} = \frac{3T^2}{3t^2} = \frac{4t^2}{t^2} \text{ (using } T = -2t)$$

$$m_B = 4$$

- Q9. A circle with centre at $(15, -3)$ is tangent to $y = \frac{x^2}{3}$ at a point in the first quadrant. The radius of the circle is equal to

(A) $5\sqrt{6}$ (B) $8\sqrt{3}$ (C) $9\sqrt{2}$ (D) $6\sqrt{5}$

Ans. D

Sol. -

- Q10. The equation of the tangent to the curve $y = x + \frac{4}{x^2}$ that is parallel to x-axis, is :

(A) $y = 0$ (B) $y = 1$ (C) $y = 2$ (D) $y = 3$

Ans. D

Sol. $y = x + \frac{4}{x^2}$

$$\frac{dy}{dx} = 1 - \frac{8}{x^3}$$

Equation of tangent is parallel to x-axis

$$\therefore \frac{dy}{dx} = 0$$

$$\Rightarrow 1 - \frac{8}{x^3} = 0 \Rightarrow x^3 = 8 \Rightarrow x = 2$$

$$\text{At, } x = 2, y = 2 + \frac{4}{4} = 3 \Rightarrow y_1 = 3$$

$\therefore$ point is $(2, 3)$

equation of tangent is :

$$y - y_1 = 0(x - x_1)$$

$$y = 3$$

Q11.

Ans. -

Sol. -

- Q12. The shortest distance between the line $y - x = 1$ and the curve $x = y^2$ is :-

(A) $\frac{3\sqrt{2}}{5}$ (B) $\frac{\sqrt{3}}{4}$ (C) $\frac{3\sqrt{2}}{8}$ (D) $\frac{2\sqrt{3}}{8}$

Ans. C

Sol. Point (t^2, t) is on the parabola $x = y^2$

Its distance from $y - x = 1$

$$d(t) = \frac{t^2 - t + 1}{\sqrt{2}}$$

$$d'(t) = \frac{1}{2}[2t - 1] = 0$$

$$t = \frac{1}{2}$$

$$d''(t) = \frac{2}{\sqrt{2}} > 0$$

$$d(t) \text{ is min at } t = \frac{1}{2}$$

Its value

$$d\left(\frac{1}{2}\right) = \frac{1}{\sqrt{2}}\left(\frac{1}{4} - \frac{1}{2} + 1\right)$$

$$d\left(\frac{1}{2}\right) = \frac{3\sqrt{2}}{8}$$

Q13. The lines tangent to the curves $y^3 - x^2y + 5y - 2x = 0$ and $x^4 - x^3y^2 + 5x + 2y = 0$ at the origin intersect at an angle θ equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Ans. D

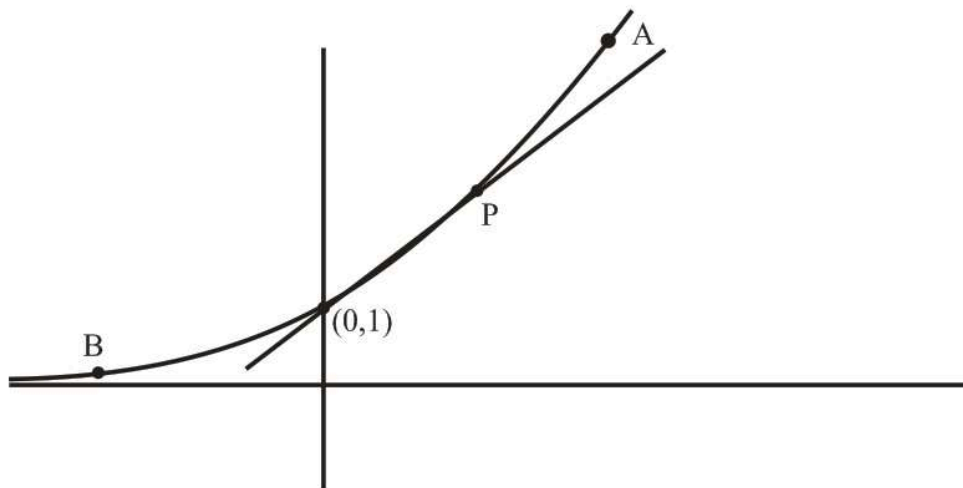
Sol. -

Q14. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$

- (A) on the left of $x = c$ (B) on the right of $x = c$ (C) at no point
(D) at all points

Ans. A

Sol.



$$P(e, e^e)$$

Slope of tangent at P

$$\frac{dy}{dx} = e^e$$

$$A(c - 1, e^{c-1})$$

$$B(c+1, e^{c+1})$$

$$m_{AB} = \frac{e^{c+1} - e^{c-1}}{2}$$

$$= \frac{e^{c+1} - e^{c-1}}{2}$$

here

$$e^c > \frac{e^{c+1} - e^{c-1}}{2}$$

hence they will intersect on left of $x = c$.

Q15. If curve $y = 1 - ax^2$ and $y = x^2$ intersect orthogonally then the value of a is

- (A) $1/2$ (B) $1/3$ (C) 2 (D) 3

Ans. B

Sol. -

Q16. If the curves $y^2 = 6x$, $9x^2 + by^2 = 16$ intersect each other at right angles, then the value of b is

- (A) 4 (B) $\frac{9}{2}$ (C) 6 (D) $\frac{7}{2}$

Ans. B

Sol.

$$y^2 = 6x$$

$$2yy' = 6$$

$$y' = \frac{3}{y_1}$$

$$9x^2 + by^2 = 16$$

$$18x + 2byy' = 0$$

$$y' = \frac{-9x_1}{by_1}$$

$$m_1 m_2 = -1$$

$$27x_1 = by_1^2 \quad \dots (i)$$

$$6x_1 = y_1^2 \quad \dots (ii)$$

$$(i) \div (ii)$$

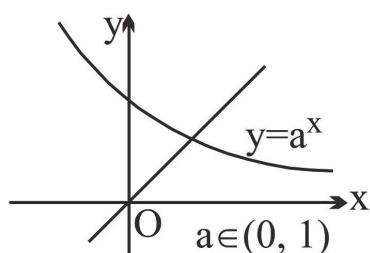
$$b = \frac{9}{2}$$

Q17. Number of positive integral values of ' a ' for which the curve $y = ax$ intersects the line $y = x$ is

- (A) 0 (B) 1 (C) 2 (D) More than 2

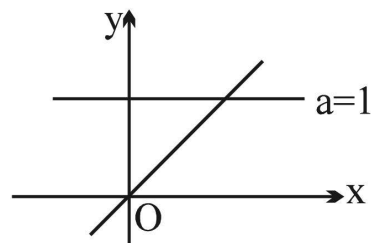
Ans. B

Sol.



for $0 < a \leq 1$ the line always cuts $y = ax$

for $a > 1$ say $a = e$



consider $f(x) = e^x - x$

$f(0)$ is the minimum

$$f'(x) = e^x - 1$$

$$f(x)|_{\min} = 1$$

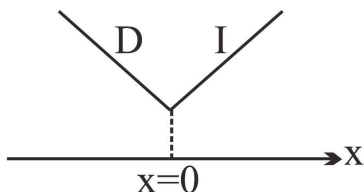
$f'(x) > 0$ for $x > 0$ and $f'(x) < 0$ for $x < 0$

$$f(x) \geq 1$$

$\therefore f(x)$ is increasing ($\uparrow$) for $x > 0$ and decreasing ($\downarrow$) for $x < 0$

$$e^x - x \geq 1$$

$y = e^x$ always lies above $y = x$ i.e. $e^x - x \geq 1$ for $a > 1$



hence never cuts $= a = (0, 1] \Rightarrow (B)$

Q18. The value of k for which the equation $e^{2x} = k\sqrt{x}$ has exactly one solution, is

- (A) $\frac{1}{\sqrt{e}}$ (B) $\sqrt{e}$ (C) $2\sqrt{e}$ (D) $\frac{2}{\sqrt{e}}$

Ans. C

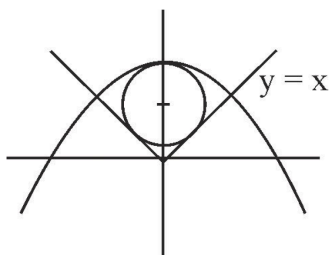
Sol. -

Q19. The radius of a circle, having minimum area, which touches the curve $y = 4 - x^2$ and the lines $y = |x|$ is

- (A) $4(\sqrt{2} - 1)$ (B) $4(\sqrt{2} + 1)$ (C) $2(\sqrt{2} + 1)$ (D) $2(\sqrt{2} - 1)$

Ans. A

Sol.



Center will be $(0, 4 - r)$

Distance of $(0, 4 - r)$ from $L : x - y = 0$

radius = r

$$\left| \frac{4-r}{\sqrt{2}} \right| = r$$

$$4-r = \sqrt{2} r$$

$$r = 4(\sqrt{2}-1)$$

Q20. At any point of a curve $\sqrt{\frac{\text{sub normal}}{\text{sub tangent}}}$ is equal to -

- (A) the abscissa of that point (B) the ordinate of that point
(C) slope of the tangent at that point (D) slope of the normal at that point

Ans. C

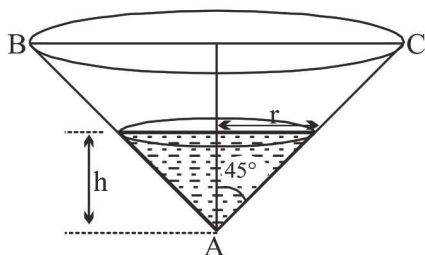
Sol. -

Q21. Water is poured at the rate of $2\text{m}^3/\text{sec}$. into a cone of semi-vertical angle 45° . The rate at which periphery of water surface changes when height of the water in the cone is 2 meter, is

- (A) 1 m/sec. (B) 2 m/sec. (C) 3m/sec. (D) 4 m/sec.

Ans. A

Sol.



We have

$$\frac{dV}{dt} = 2 \Rightarrow \frac{d}{dt} \left(\frac{1}{3} \pi r^3 \right) = 2 \quad [\text{Here } r = h, \text{ as } \theta = 45^\circ]$$

$$\Rightarrow \pi r^2 \frac{dr}{dt} = 2 \Rightarrow \frac{dr}{dt} = \frac{2}{\pi r^2} \quad \dots\dots(1)$$

Now, perimeter = $2\pi r = p$ (let)

$$\Rightarrow \frac{d}{dt} (2\pi r) = 2\pi \left(\frac{2}{\pi r^2} \right) = \frac{4}{\pi r^2} \quad \dots\dots(2) \quad (\text{using equation (1)})$$

When $h = 2$ meters $\Rightarrow r = 2$ meters

$$\text{Hence } \frac{dp}{dt} = \frac{4}{4} = 1 \text{ m/sec.}$$

Q22. A point is moving along the curve $y^3 = 27x$. The interval in which the abscissa changes at slower rate than ordinate, is

- (A) $(-3, 3)$ (B) $(-\infty, \infty)$ (C) $(-1, 1)$ (D) $(-\infty, -3) \cup (3, \infty)$

Ans. C

Sol. -

Q23. A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate of which the thickness of ice decreases, is -

- (A) $\frac{1}{36\pi} \text{ cm/min}$ (B) $\frac{1}{18\pi} \text{ cm/min}$ (C) $\frac{1}{54\pi} \text{ cm/min}$ (D) $\frac{5}{6\pi} \text{ cm/min}$

Ans. A

Sol. -

Q24. The function $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is

- (A) increasing in its domain (B) decreasing in its domain
(C) decreasing in $(-\infty, 0)$ and increasing in $(0, \infty)$
(D) increasing in $(-\infty, 0)$ and decreasing in $(0, \infty)$

Ans. D

Sol. -

Q25. If $f(x) = xe^{x(1-x)}$ then $f(x)$ is

- (A) Increasing on $\left[-\frac{1}{2}, 1\right]$ (B) Decreasing on $\mathbb{R}$ (C) Increasing on $\mathbb{R}$
(D) Decreasing on $\left[-\frac{1}{2}, 1\right]$

Ans. A

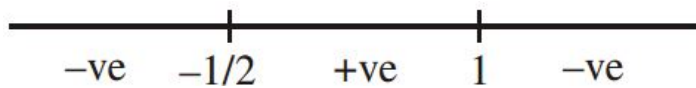
Sol. We have,

$$f'(x) = e^{x(1-x)} + x \cdot e^{x(1-x)} \cdot (1-2x)$$

$$= e^{x(1-x)} [1 + x(1-2x)]$$

$$= e^{x(1-x)} \cdot [-2x^2 + x + 1]$$

Now, the sign-scheme for $-2x^2 + x + 1$ is



$\therefore f'(x) \geq 0$ if $x \in \left[-\frac{1}{2}, 1\right]$ $e^{x(1-x)}$ is always positive

So, $f(x)$ is increasing on $\left[-\frac{1}{2}, 1\right]$

Q26. If $f(x) = xe^{x(1-x)}$, then $f(x)$ is

(A) increasing on $\left(-\frac{1}{2}, 1\right)$ (B) decreasing on $\left[-\frac{1}{2}, 1\right]$ (C) increasing on $\mathbb{R}$

(D) decreasing on $\mathbb{R}$

Ans. A

Sol. -

Q27. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in-

(A) $(\pi/4, \pi/2)$ (B) $(-\pi/2, \pi/4)$ (C) $(0, \pi/2)$ (D) $(-\pi/2, \pi/2)$

Ans. B

Sol. -

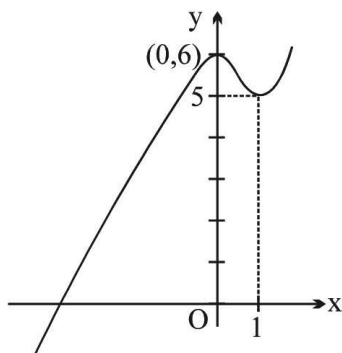
Q28. The function $f: [a, \infty) \rightarrow \mathbb{R}$ where $\mathbb{R}$ denotes the range corresponding to the given domain, with rule

$f(x) = 2x^3 - 3x^2 + 6$, will have an inverse provided

(A) $a \geq 1$ (B) $a \geq 0$ (C) $a \leq 0$ (D) $a \leq 1$

Ans. A

Sol.



$$f(x) = 2x^3 - 3x^2 + 6$$

$$f'(x) = 6x^2 - 6x = 6(x^2 - x) = 0 \quad \text{gives } x = 0 \text{ or } x = 1$$

for inverse to exist function must be one one onto

hence Domain is $[1, \infty)$

Hence $a \geq 1]$

Q29. Let $f(x) = x - \frac{1}{x}$ then which one of the following statement is true

(A) Function is invertible if defined from $\mathbb{R} - \{0\} \rightarrow \mathbb{R}$.

(B) $f(x_1) > f(x_2)$, $\forall x_1 > x_2$ and $x_1, x_2 \neq 0$

(C) Graph of the function has exactly one asymptote.

(D) Function is one-one in every continuous interval $[a, b]$ defined on one side of origin.

Ans. D

Sol. -

Q30. If the function $f(x) = 2x^2 - kx + 5$ is increasing in $[1, 2]$, then 'k' lies in the interval

(A) $(-\infty, 4)$ (B) $(-4, \infty)$ (C) $(-\infty, 8)$ (D) $(8, \infty)$

Ans. A

Sol. Given, $f(x) = 2x^2 - kx + 5$ Now $f'(x) > 0$ for $1 \leq x \leq 2$

$$f'(x) = 4x - k, f''(x) = 4 \Rightarrow f''(x) > 0 \text{ for all } x \in [1, 2]$$

$$\Rightarrow f'(x) \text{ is an increasing function in } [1, 2] \Rightarrow f'(x) > 0$$

$$\therefore f'(1) \text{ is the least value of } f'(x) \text{ in } [1, 2] \Rightarrow f'(1) > 0 \Rightarrow 4 - k > 0$$

$$\Rightarrow k < 4 \Rightarrow k \in (-\infty, 4)$$

Q31. Consider the following behaviours of function in $(-1, 1)$

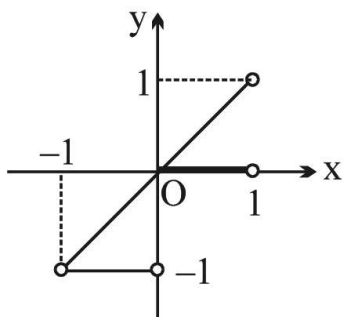
I. Increasing II. Decreasing III. Continuity IV. Derivability

Which one of the following four functions exhibits atleast three of the four mentioned above?

- (A) $\max \{x, x^3\}$ (B) $\max \{x, x^2\}$ (C) $\max \{x, |x|\}$ (D) $\max \{x, [x]\}$

Ans. D

Sol. $\max. \{x, [x]\}$ in $(-1, 1)$ is x as $[x] \leq x$



Q32. Consider the following statements S and R

S : Both $\sin x$ & $\cos x$ are decreasing functions in the interval $(\pi/2, \pi)$.

R : If a differentiable function decreases in an interval (a, b) , then its derivative also decreases in (a, b) .

Which of the following is true ?

- (A) both S and R are wrong
 (B) both S and R are correct, but R is not the correct explanation for S
 (C) S is correct and R is the correct explanation for S
 (D) S is correct and R is wrong.

Ans. D

Sol. -

Q33. Number of critical points of the function, $f(x) =$

$$\frac{2}{3}\sqrt{x^3} - \frac{x}{2} + \int_1^x \left(\frac{1}{2} + \frac{1}{2}\cos 2t - \sqrt{t} \right) dt$$

which lie in the interval $[-2\pi, 2\pi]$ is :

- (A) 2 (B) 4 (C) 6 (D) 8

Ans. B

Sol. -

Q34. Let $f(x) = \begin{cases} x^{3/5} & \text{if } x \leq 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$

then the number of critical points on the graph of the function is

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. -

Q35.

$$\text{Let a function } f \text{ be defined as } f(x) = \begin{cases} \frac{|x-1|}{x^2+1} & \text{if } x > -1 \\ x^2 & \text{if } x \leq -1 \end{cases}$$

Then the number of critical point(s) on the graph of this function is/are :

- (A) 4 (B) 3 (C) 2 (D) 1

Ans. A

Sol. -

Q36. Suppose that f is a polynomial of degree 3 and that $f''(x) \neq 0$ at any of the stationary point. Then

- (A) f has exactly one stationary point. (B) f must have no stationary point.
(C) f must have exactly 2 stationary points. (D) f has either 0 or 2 stationary points.

Ans. D

Sol. -

Q37. The set of all values of 'a' for which the function ,

$f(x) = (a^2 - 3a + 2) \left(\cos^2 \frac{x}{4} - \sin^2 \frac{x}{4} \right) + (a-1)x + \sin 1$ does not possess critical points is:

- (A) $[1, \infty)$ (B) $(0, 1) \cup (1, 4)$ (C) $(-2, 4)$ (D) $(1, 3) \cup (3, 5)$

Ans. B

Sol. -

Q38. For all $x \in (0, 1)$

- (A) $e^x < 1 + x$ (B) $\log_e(1+x) < x$ (C) $\sin x > x$ (D) $\log_e x > x$

Ans. B

Sol. $x \in (0, 1)$

(A) $e^x < 1 + x$

$$1 + \frac{x}{L_1} + \frac{x^2}{L_2} + \frac{x^3}{L_3} + \dots < 1 + x$$

$$\frac{x}{L_2} + \frac{x^3}{L_3} + \frac{x^3}{L_3} + \dots < 0$$

Incorrect

(B) $\ln(1+x) \dots < x$

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \dots < 0$$

Correct

(C) $\sin x > x$

$$x - \frac{x^3}{L_3} + \frac{x^5}{L_5} \dots > 0$$

$$-\frac{x^3}{L_3} + \frac{x^5}{L_5} > 0$$

Incorrect

$$x > e^x$$

Incorrect

Q39. The true set of real values of x for which the function, $f(x) = x \ln x - x + 1$ is positive is

- (A) $(1, \infty)$ (B) $(1/e, \infty)$ (C) $[e, \infty)$ (D) $(0, 1)$ and $(1, \infty)$

Ans. D

Sol. -

Q40. The set of all x for which $\ln(1+x) \leq x$ is equal to

- (A) $x > 0$ (B) $x > -1$ (C) $-1 < x < 0$ (D) null set

Ans. B

Sol. -

Q41. The equation $\sin x + x \cos x = 0$ has at least one root in

- (A) $\left(-\frac{\pi}{2}, 0\right)$ (B) $(0, \pi)$ (C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

Ans. B

Sol. -

Q42. Let $f(x) = (x-4)(x-5)(x-6)(x-7)$ then,

- (A) $f(x) = 0$ has four roots (B) Three roots of $f(x) = 0$ lie in $(4, 5) \cup (5, 6) \cup (6, 7)$
(C) The equation $f(x) = 0$ has only one real root
(D) Three roots of $f(x) = 0$ lie in $(3, 4) \cup (4, 5) \cup (5, 6)$

Ans. B

Sol. -

Q43. Let a, b, c, d are non-zero real numbers such that $6a + 4b + 3c + 3d = 0$, then the equation $ax^3 + bx^2 + cx + d = 0$ has

- (A) atleast one root in $[-2, 0]$ (B) atleast one root in $[0, 2]$
(C) atleast two roots in $[-2, 2]$ (D) no root in $[-2, 2]$

Ans. B

Sol. We have $ax^3 + bx^2 + cx + d = 0$

$$\text{Let } f(x) = \frac{ax^3}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + dx + e$$

$$\therefore f(0) = e$$

$$f(2) = 4a + \frac{8b}{3} + 2c + 2d + e = \frac{(12a + 8b + 6c + 6d)}{3} + e = \frac{2}{3}(6a + 4b + 3c + 3d) + e = 0 + e$$

$$\Rightarrow f(2) = e$$

$\therefore$ By Rolle's theorem, there exist atleast one value of $x \in (0, 2)$ such that $f'(x) = 0$

$\Rightarrow$ The equation $ax^3 + bx^2 + cx + d = 0$ has atleast one real root in $[0, 2]$.

Q44. If the equation $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x = 0$; $a_1 \neq 0, n \geq 2$, has a positive root $x = \alpha$, then the equation

$na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ has a positive root, which is -

- (A) greater than α (B) smaller than α (C) greater than or equal to α
(D) equal to α

Ans. B

Sol. Let $f(x) = anx^3 + a_{n-1}x^{n-1} + \dots + a_1x$

It is given that one root is '0' and positive root is $x = \alpha$

$$\text{Let } g(x) = \int (na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1) dx$$

$$g(x) = anx^3 + a_{n-1}x^{n-1} + \dots + a_1x + K$$

$g(x)$ is continuous in $(0, \alpha)$ also differentiable and

$$g(0) = K$$

$$g(\alpha) = 0 + K$$

hence Rolle's theorem can be applied on $g(x)$ in $(0, \alpha)$

hence

$$g'(x) = nanx^{n-1} + (n-1)a_{n-1}x^{n-2} + \dots + a_1 = 0$$

will have a root in interval $(0, \alpha)$

hence correct option is (β) .

Q45.

Ans. -

Sol. -

Q46. Let $f(x) = x + \sqrt{x}$ on $[1, 4]$. The mean value theorem says that there must be some number 'c' between 1 and 4 so that $f'(c)$ is equal to the average slope of $f(x)$ on $[1, 4]$. the number 'c' must be

(A) $\frac{5}{2}$ (B) $\frac{9}{4}$ (C) $\frac{11}{4}$ (D) 3

Ans. B

Sol. $f'(c) = \frac{f(4) - f(1)}{4 - 1}$

$$1 + \frac{1}{2\sqrt{c}} = \frac{6 - 2}{3} = \frac{4}{3} \Rightarrow \frac{1}{2\sqrt{c}} = \frac{1}{3} \Rightarrow \sqrt{c} = \frac{3}{2} \Rightarrow c = \frac{9}{4}$$

Q47. Let f be differentiable for all x . If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$, then -

(A) $f(6) \geq 8$ (B) $f(6) < 8$ (C) $f(6) < 5$ (D) $f(6) = 5$

Ans. A

Sol. By LMVT (LMVT)

$$\frac{f(6) - f(1)}{6 - 1} = f'(x) \quad x \in (1, 6)$$

$$\text{and } f'(x) \geq 2 \Rightarrow \frac{f(6) - f(1)}{6 - 1}$$

$$\Rightarrow \frac{f(6) + 2}{5} \geq 2 \Rightarrow f(6) \geq 8$$

Q48. A value of C for which the conclusion of Mean Value Theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is-

(A) $2\log_3 e$ (B) $\frac{1}{2}\log_e 3$ (C) $\log_3 e$ (D) $\log_e 3$

Ans. A

Sol. -

Q49. Difference between the greatest and the least values of the function $f(x) = x(\ln x - 2)$ on $[1, e^2]$ is

- (A) 2 (B) e (C) e^2 (D) 1

Ans. B

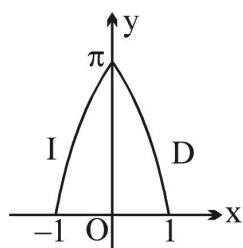
Sol. -

Q50. Which one of the following statements does not hold good for the function $f(x) = \cos^{-1}(2x^2 - 1)$?

- (A) f is not differentiable at $x = 0$ (B) f is monotonic (C) f is even
(D) f has an extremum

Ans. B

Sol. $\cos^{-1}(2x^2 - 1) = \begin{cases} 2\cos^{-1}x & \text{for } x \in [0,1] \\ 2\pi - 2\cos^{-1}x & \text{for } x \in [-1,0] \end{cases}$



f is increasing in $(-1, 0)$ and decreasing in $(0, 1)$

$\Rightarrow$ non monotonic $\Rightarrow$ (B) is wrong

Q51. The minimum value of $f(x, y) = x^2 - 4x + y^2 + 6y$ when x and y are subjected to the restrictions $0 \leq x \leq 1$ and $0 \leq y \leq 1$, is

- (A) -1 (B) -2 (C) -3 (D) -5

Ans. C

Sol. We have $f(x, y) = x^2 + y^2 - 4x + 6y$

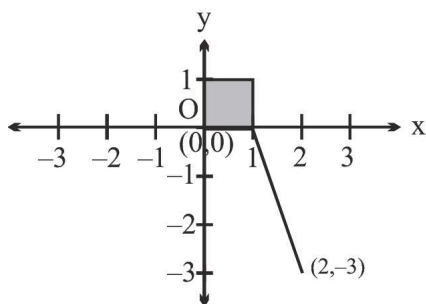
Let $(x, y) = (\cos \theta, \sin \theta)$, then $\theta \in [0, \frac{\pi}{2}]$ and

$$f(x, y) = f(\theta) = \cos^2 \theta + \sin^2 \theta - 4 \cos \theta + 6 \sin \theta$$

$$f'(\theta) = 6 \cos \theta + 4 \sin \theta > 0 \quad \forall \theta \in [0, \frac{\pi}{2}]$$

$\therefore f'(\theta)$ is strictly increasing in $[0, \frac{\pi}{2}]$

$\therefore f(\theta) \min = f(0) = 1 - 4 + 0 = -3$ Ans.]



Alternatively: $f(x, y) = (x - 2)^2 + (y + 3)^2 - 13$.

So, to minimise f over the unit square is the same as minimising the distance from a point in this square to the point of coordinates $(2, -3)$.

The closest point in the square to $(2, -3)$ is then $(1, 0)$.

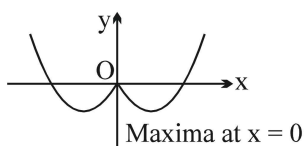
So $f(1, 0) = -3$ must be the minimum.]

Q52. Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at $x = 0$, $f(x)$ has

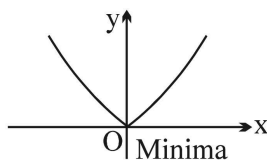
- (A) a maxima whenever $a > 0, b > 0$ (B) a maxima whenever $a > 0, b < 0$
 (C) minima whenever $a > 0, b > 0$
 (D) neither a maxima nor minima whenever $a > 0, b < 0$

Ans. A

Sol. for $a > 0, b > 0$



for $a > 0, b < 0$



Q53. If a function f is continuous for all x and if f has a relative maximum at $(-1, 4)$ and a relative minimum at $(3, -2)$, then which of the following statements must be true?

- (A) The graph of f has a point of inflection somewhere between $x = -1$ and $x = 3$.
 (B) $f'(-1) = 0$. (C) The graph of f has a horizontal tangent line at $x = 3$.
 (D) The graph of f intersect both co-ordinate axes.

Ans. D

Sol. Because f is continuous for all x , the intermediate value theorem implies that the graph of f must intersect the x -axis. The graph must also intersect the y -axis since f is defined for all x , in particular, at $x = 0$. As f need not be differentiable. Hence A, B and C need not be correct. **Ans.**

Q54. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals

- (A) $1/2$ (B) 3 (C) 1 (D) 2

Ans. D

Sol. -

Q55. If p and q are positive real numbers such that $p^2 + q^2 = 1$, then the maximum value of $(p + q)$ is-

- (A) 2 (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

Ans. D

Sol. Using A.M. $\geq$ G.M.

$$\frac{p^2 + q^2}{2} \geq p \cdot q$$

$$\Rightarrow pq \leq \frac{1}{2}$$

$$\Rightarrow (p + q)^2 = p^2 + q^2 + 2pq$$

$$\Rightarrow (p + q) \leq \sqrt{2}$$

Q56. Suppose the cubic $x^3 - px + q$ has three distinct real roots where $p > 0$ and $q > 0$. Then which one of the following holds ?

(A) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$

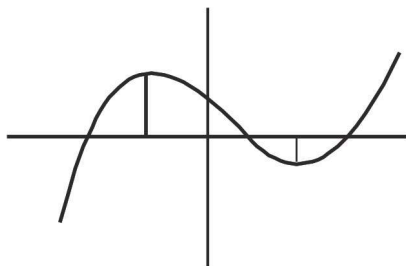
(B) The cubic has minima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

(C) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

(D) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$

Ans. D

Sol. Graph of $y = x^3 - px + q$ cuts x-axis at three distinct points



$$\frac{dy}{dx} = 0 \Rightarrow x = \pm \sqrt{\frac{p}{3}}$$

Maxima at $x = -\sqrt{\frac{p}{3}}$, minima at $x = \sqrt{\frac{p}{3}}$

Q57.

Ans. -

Sol. -

Q58. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} k-2x, & \text{if } x \leq -1 \\ 2x+3, & \text{if } x > -1 \end{cases}$ If f has a local minimum at $x = -1$, then a possible value of k is :

- (A) 1 (B) 0 (C) $-\frac{1}{2}$ (D) -1

Ans. D

Sol. f has a local minimum at $x = -1$

$$\therefore \lim_{x \rightarrow -1} f(x) \geq f(-1)$$

$$k + 2 \leq 1$$

$$k \leq -1$$

$$\therefore k = -1$$

Directions : Questions number 86 to 90 are Assertion - Reason type questions. Each of these questions contains two statements :

Statement-1 (Assertion) and

Statement-2 (Reason).

Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select the correct choice.

Q59. For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t} \sin t dt$. Then f has :-

- (A) local minimum at π and local maximum at 2π
 (B) local maximum at π and local minimum at 2π (C) local maximum at π and 2π
 (D) local minimum at π and 2π

Ans. B

Sol. $f'(x) = \sqrt{x} \sin x$

$f'(\pi)$ & $f'(2\pi)$ are 0.

$$f'(x) \quad \begin{array}{c} + \quad \quad - \quad \quad + \\ \hline \pi \quad \quad 2\pi \end{array}$$

π local maximum at $x = \pi$ and local minimum at $x = 2\pi$

Q60. If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log |x| + \beta x^2 + x$ then :

- (A) $\alpha = 2, \beta = -\frac{1}{2}$ (B) $\alpha = 2, \beta = -\frac{1}{2}$ (C) $\alpha = -6, \beta = \frac{1}{2}$ (D) $\alpha = -6, \beta = -\frac{1}{2}$

Ans. A

Sol. $f(x) = \alpha \ln |x| + \beta x^2 + x$

$$(1) f'(x) = \frac{\alpha}{x} + 2\beta x + 1$$

$$= \frac{2\beta x^2 + x + l}{x}$$

Since $x = -1, 2$ are extreme points $f'(x) = 0$ at these points.

$$\text{Hence } 2\beta - 1 + \alpha = 0$$

$$8\beta + 2\alpha = 0$$

$$-6\beta - 3 = 0 \Rightarrow \beta = -\frac{1}{2} \quad \& \quad \alpha = 2$$

Q61. Let $f(x)$ be a polynomial of degree four having extreme values at $x = 1$ and $x = 2$. If $\lim_{x \rightarrow \infty} \left[1 + \frac{f(x)}{x^2} \right] = 3$, then $f(2)$ is equal to :

- (A) -8 (B) -4 (C) 0 (D) 4

Ans. C

$$\begin{aligned} \text{Sol. } f'(1) &= 0 & f(x) &= ax^4 + bx^3 + 2x^2 \\ f'(2) &= 0 & f'(1) &= 0 \\ & & 4a + 3b + 4 &= 0 \end{aligned}$$

$$f(x) = \frac{x^4}{2} - x^3 + 2x^2 \quad f'(1) = 0$$

$$f(2) = 8 - 16 + 8 = 0 \quad \frac{32a + 12b + 8 = 0}{a = \frac{1}{2}, b = -2}$$

Q62. Let $f(x) = x^2 + \frac{1}{x^2}$ and, $g(x) = x - \frac{1}{x}, x \in R - \{-1, 0, 1\}$. If $h(x) = \frac{f(x)}{g(x)}$, then the local minimum value of $h(x)$ is

- (A) $-2\sqrt{2}$ (B) $2\sqrt{2}$ (C) 3 (D) -3

Ans. B

$$\text{Sol. } h(x) = \frac{x^2 + \frac{1}{x^2}}{x - \frac{1}{x}}$$

$$h(x) = \frac{t^2 + 2}{t} = t + \frac{2}{t} \text{ where } t = x - \frac{1}{x}$$

$$h'(x) = \left(1 - \frac{2}{t^2} \right) \left(\frac{dt}{dx} \right)$$

$$h'(x) = \left(\frac{t^2 - 2}{t} \right) \left(1 + \frac{1}{x^2} \right)$$

Minima at $t = \sqrt{2}$

$$h(x)|_{Min} = \frac{2+2}{\sqrt{2}} = 2\sqrt{2}$$

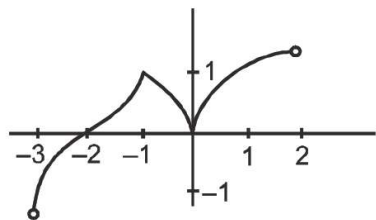
Q63. The total number of local maxima and local minima of the function

$$f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{\frac{2}{3}}, & -1 < x < 2 \end{cases} \text{ is}$$

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol.
$$f(x) = \begin{cases} (2+x)^2, & -3 < x \leq -1 \\ x^{\frac{2}{3}}, & -1 < x < 2 \end{cases}$$



one point of local maxima at $x = -1$

one point of local minima at $x = 0$

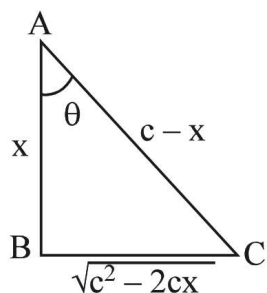
$\therefore$ (C) is the correct option

Q64. The sum of lengths of the hypotenuse and another side of a right angled triangle is given. The area of the triangle will be maximum if the angle between them is :

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Ans. C

Sol.



$$A = \frac{x\sqrt{c^2 - 2cx}}{2}$$

$$f(x) = 4A^2 = x^2(c^2 - 2cx)$$

Q65. The length of the shortest path that begins at the point (2, 5), touches the x-axis and then ends at a point on the circle $x^2 + y^2 + 12x - 20y + 120 = 0$, is

- (A) 13 (B) $4\sqrt{10}$ (C) 15 (D) $6 + \sqrt{89}$

Ans. A

Sol. Circle with centre $(-6, 10)$ and radius $= \sqrt{36 + 100 - 120} = 4$

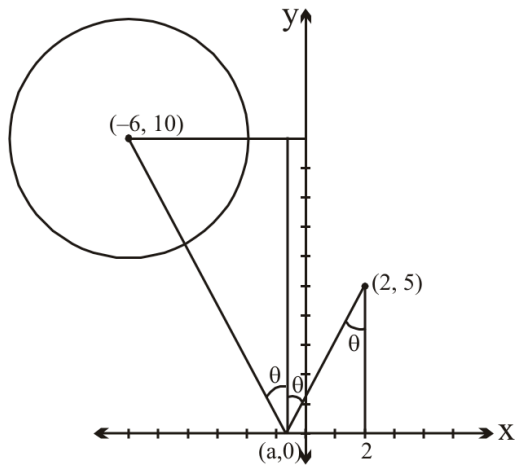
Now let $(a, 0)$ be a point on the x-axis.

If y is the distance from A to P and P to M

$$y = \sqrt{(a-2)^2 + 25} + \sqrt{(a+6)^2 + 100} - 4$$

$$\frac{dy}{da} = \frac{2(a-2)}{2\sqrt{(a-2)^2 + 25}} + \frac{2(a+6)}{2\sqrt{(a+6)^2 + 100}}$$

$\frac{dy}{da}$ can be zero only if $a-2 > 0$



and $a+6 < 0$ not possible

or $a-2 < 0$ and $a+6 > 0$

hence $a \in (-6, 2)$

solving $\frac{dy}{da} = 0$, gives $a = 10$ (rejected) or $a = -2/3$

$$\text{hence } y_{\min} = \sqrt{\frac{64}{9} + 25} + \sqrt{\frac{256}{9} + 100} - 4$$

$$= \frac{17}{3} + \frac{\sqrt{1156}}{3} - 4 = \frac{17}{3} + \frac{34}{3} - 4 = 17 - 4 = 13 \text{ Ans.}$$

$$\text{A loter : } \frac{a+2}{5} = \frac{6-a}{10} \Rightarrow 10a + 20 = 30 - 5a$$

$$\Rightarrow 15a = 10 \Rightarrow a = \frac{2}{3}$$

Hence $P\left(\frac{-2}{3}, 0\right)$ for find $y_{\min}$.

Q66. A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x . The maximum area enclosed by the park is –

$$(A) \sqrt{\frac{x^3}{8}} \quad (B) \frac{1}{2}x^2 \quad (C) \pi x^2 \quad (D) \frac{3}{2}x^2$$

Ans. B

Sol. -

Q67. A right triangle is drawn in a semicircle of radius $\frac{1}{2}$ with one of its legs along the diameter. The maximum area of the triangle is

(A) $\frac{1}{4}$ (B) $\frac{3\sqrt{3}}{32}$ (C) $\frac{3\sqrt{3}}{16}$ (D) $\frac{1}{8}$

Ans. B

Sol. -

Q68. Two sides of a triangle are to have lengths 'a' cm & 'b' cm. If the triangle is to have the maximum area, then the length of the median from the vertex containing the sides 'a' and 'b' is

(A) $\frac{1}{2}\sqrt{a^2 + b^2}$ (B) $\frac{2a + b}{3}$ (C) $\sqrt{\frac{a^2 + b^2}{2}}$ (D) $\frac{a + 2b}{3}$

Ans. A

Sol. -

Q69. A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is :

(A) $\frac{9}{2}$ (B) $\frac{9}{7}$ (C) $\frac{7}{9}$ (D) $\frac{2}{9}$

Ans. D

Sol. $V = \frac{4}{3}\pi r^3$

Initially $r = 4500 \pi r = r_0$

$$4500\pi = \frac{4}{3}\pi r_0^3 \Rightarrow \boxed{r_0 = 15m}$$

Now $\frac{dV}{dt} = \frac{4}{3}\pi(3r^2) \frac{dr}{dt}$

$$-72\pi = 4\pi r^2 \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{-18}{r^2} \dots\dots (i)$$

$$\int r^2 dr = - \int 18 dt \Rightarrow \frac{r^3}{3} = -18t + C$$

At $t = 0$, $r = 15$ m

So, $\frac{(15)^3}{3} = -18(0) + C \Rightarrow C = 1125$

$$\Rightarrow r^3 = -54t + 3375 \dots\dots(ii)$$

At time $t = 49$ min $r = 9$ m

from eq. (i)

$$\left(\frac{dr}{dt}\right)_{t=49} = \frac{-18}{(9)^2} = -\frac{2}{9}$$

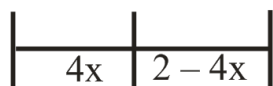
(Negative sign shows decrement in radii)

Q70. A wire of length 2 units is cut into two parts which are bent respectively to form a square of side = x units and a circle of radius = r units. If the sum of the areas of the square and the circle so formed is minimum, then :

- (A) $x = 2r$ (B) $2x = r$ (C) $2x = (\pi + 4)r$ (D) $(4 - \pi)x = \pi r$

Ans. A

Sol.



$$2\pi r = (2 - 4x)$$

$$A = x^2 + \pi \left(\frac{2 - 4x}{2\pi} \right)^2$$

$$\frac{dA}{dx} = 0$$

$$2x = \frac{2}{\pi}(2 - 4x)$$

$$2x = \frac{2(2\pi r)}{\pi}$$

$$x = 2r$$

Q71.

Ans. -

Sol. -

Q72.

Ans. -

Sol. -

Q73. The function $f(x) = 2x^3 + \alpha x^2 + \beta x + \gamma$, where $\alpha, \beta, \gamma \in \mathbb{R}$ has local minimum at $P(\log_3 t^2, f(\log_3 t^2))$ and local maximum at $Q(\log_3 t, f(\log_3 t))$. If $R(\frac{5}{2}, f(\frac{5}{2}))$ is the point of inflection, then 't' is equal to -

- (A) $3^{\frac{2}{5}}$ (B) $3^{\frac{1}{5}}$ (C) $3^{\frac{5}{3}}$ (D) $3^{\frac{3}{5}}$

Ans. C

Sol. $f'(x) = 6x^2 + 2\alpha x + \beta \Rightarrow f'(\log_3 t^2) = 0$ and $(\log_3 t) = 0$

Now $\log_3 t^2 + \log_3 t = -\frac{2\alpha}{6}$ (sum of the roots)

$$\Rightarrow 3\log_3 t = \frac{-\alpha}{3} \dots\dots\dots(i)$$

Also, $f''(x) = 12x + 2\alpha$

$$f''\left(\frac{5}{2}\right) = 0 \Rightarrow 12\left(\frac{5}{2}\right) + 2\alpha \Rightarrow \alpha = -15$$

$$3\log_3 t = \frac{15}{3} \Rightarrow t = 3^{\frac{5}{3}}$$

Q74. If the point (1, 3) serves as the point of inflection of the curve $y = ax^3 + bx^2$ then the value of 'a' and 'b' are:

- (A) $a = 3/2$ & $b = -9/2$ (B) $a = 3/2$ & $b = 9/2$ (C) $a = -3/2$ & $b = -9/2$
 (D) $a = -3/2$ & $b = 9/2$

Ans. D

Sol. -

Exercise J2

Q1.

Ans. -

Sol. -

Q2. Find the point of intersection of the tangents drawn to the curve $x^2y = 1 - y$ at the points where it is intersected by the curve $xy = 1 - y$.

Ans. (0,1)

Sol. Points of intersection of curve $x^2y = xy$ are

$$(0, 1), (1, \frac{1}{2})$$

The equation of tangent at (0, 1)

$$2xy + x^2y = -y'$$

$$y' = 0$$

$$\text{equation is } y - 1 = 0 \Rightarrow y = 1$$

equation of tangent at $(1, \frac{1}{2})$ is $2xy + x^2y = -y'$

$$y - \frac{1}{2} = -\frac{1}{2}(x - 1)$$

$$\text{Put } y = 1 \quad \frac{1}{2} = -\frac{x}{2} + \frac{1}{2}$$

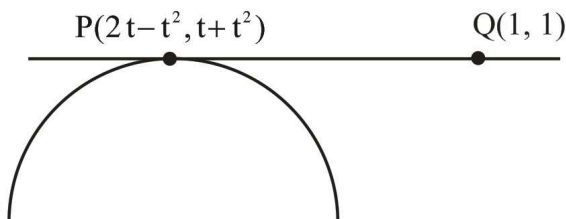
$$x = 0$$

Point of intersection of tangent is (0, 1)

Q3. Find all the lines that pass through the point (1, 1) and are tangent to the curve represented parametrically as $x = 2t - t^2$ and $y = t + t^2$.

Ans. $x=1$ when $t=1, m \rightarrow \infty$; $5x-4y=1$ if $t \neq 1, t=1/3$

Sol.



By ultimate concept

$$\frac{\frac{dy}{dt}}{\frac{dx}{dt}} = m_{PQ}$$

$$\frac{1+2t}{2-2t} = \frac{t^2+t-1}{2t-t^2-1}$$

$$3t^2 - 4t + 1 = 0$$

$$3t^2 - 4t + 1 = 0$$

$$3t(t-1) - 1(t-1) = 0$$

$$3(t-1)(t-1) = 0$$

$$t = 1, \frac{1}{3}$$

$$P(1, 1) \text{ and } Q(1, 1)$$

tangent

$$x = 1$$

$$t = \frac{1}{3}$$

$$P\left(\frac{5}{9}, \frac{4}{9}\right) \text{ and } Q(1, 1)$$

tangent

$$5x - 4y = 1$$

Q4. A function is defined parametrically by the equations

$$f(t) = x = \begin{cases} 2t + t^2 \sin \frac{1}{t} & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases} \quad \text{and } g(t) = y = \begin{cases} \frac{1}{t} \sin t^2 & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases}$$

Find the equation of the tangent and normal at the point for $t = 0$ if exist.

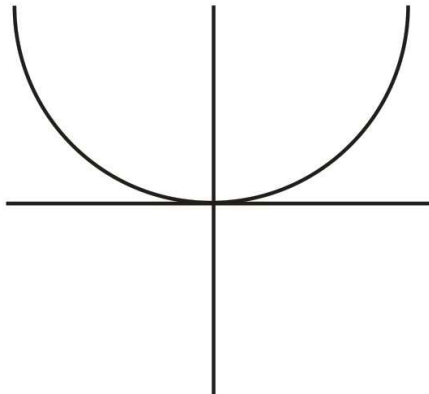
Ans. T : $x - 2y = 0$; N : $2x + y = 0$

Sol. -

Q5. Prove that the segment of the normal to the curve $x = 2a \sin t + a \sin t \cos^2 t$; $y = -a \cos^3 t$ contained between the co-ordinate axes is equal to $2a$.

Ans. -

Sol.



$$P(2a \sin t + a \sin t \cos^2 t, -a \cos^3 t)$$

Equation of normal

$$y + a \cos^3 t = mN(x - 2a \sin t - a \sin t \cos^2 t)$$

$$\frac{dy}{dx} = mT = \frac{+3a \cos^2 t \sin t}{2a \cos t + a \cos^3 t - 2a \sin^2 t + \cos t}$$

$$mT = \frac{3a \sin t \cos t}{a(2 + \cos^2 t - 2 \sin^2 t)}$$

$$mT = \frac{\sin t}{\cos t}$$

$$mN = \frac{-\cos t}{\sin t}$$

Equation of normal

$$(y + a \cos^3 t) = \frac{-\cos t}{\sin t}(x - 2a \sin t - a \sin t \cos^2 t)$$

$$y \sin t + a \cos^3 t + \sin t = -\cot t(x - 2a \sin t - a \sin t \cos^2 t)$$

X.I. of normal

put $y = 0$

$$a \cos^3 t + \sin t = -\cos t(x - 2a \sin t - a \sin t \cos^2 t)$$

$$-a \cos^2 t \sin t = x - 2a \sin t - a \sin t \cos^2 t$$

$$x = 2a \sin t + a \sin t + \cos^2 t$$

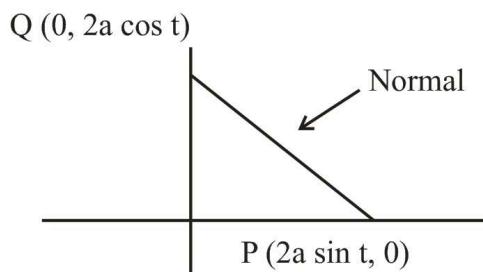
$$x = 2a \sin t$$

X.I. of normal

put $x = 0$

$$y \sin t + a \cos^3 t \sin t = +2a \sin t \cos t + a \sin t \cos^3 t$$

$$y = 2a \cos t$$



$$PQ = \sqrt{4a^2 \sin^2 t + 4a^2 \cos^2 t}$$

$$= \sqrt{4a^2}$$

$$PQ = 2a \quad \quad \quad 1 + P$$

Q6. Determine a differentiable function $y = f(x)$ which satisfies $f'(x) = [f(x)]^2$ and $f(0) = -\frac{1}{2}$. Find also the equation of the tangent at the point where the curve crosses the y-axis.

Ans. $-\frac{1}{x+2}; x-4y=2$

Sol. $\frac{dy}{dx} = y^2$
 $\int \frac{dy}{dx} = \int dx$

$$-\frac{1}{y} = x + c$$

$$\frac{-1}{f(x)} = x + c$$

$$x = 0$$

$$c = 2$$

$$-\frac{1}{f(x)} = x + 2$$

$$f(x) = \frac{-1}{(x+2)}$$

when curve cross y-axis $x = 0$

$$y = \frac{-1}{0+2} = \frac{-1}{2}$$

$$\text{Point } \left(0, \frac{-1}{2}\right)$$

$$\frac{dy}{dx} + \frac{1}{(x+2)^2} = \frac{1}{4}$$

$$y + 2 = x$$

$$x - 4y - 2 = 0$$

Q7.

Ans. -

Sol. -

Q8.

Ans. -

Sol. -

Q9.

Ans. -

Sol. -

- Q10. Tangent at a point P_1 [other than $(0, 0)$] on the curve $y = x^3$ meets the curve again at P_2 . The tangent at P_2 meets the curve at P_3 & so on. Show that the abscissae of $P_1, P_2, P_3, \dots, P_n$, form a GP. Also find the ratio $\frac{\text{area}(P_1P_2P_3)}{\text{area}(P_2P_3P_4)}$.

Ans. $1/16$

Sol. -

Q11.

Ans. -

Sol. -

- Q12. The graph of a certain function f contains the point $(0, 2)$ and has the property that for each number ' p ' the line tangent to $y = f(x)$ at $(p, f(p))$ intersect the x-axis at $p + 2$. Find $f(x)$.

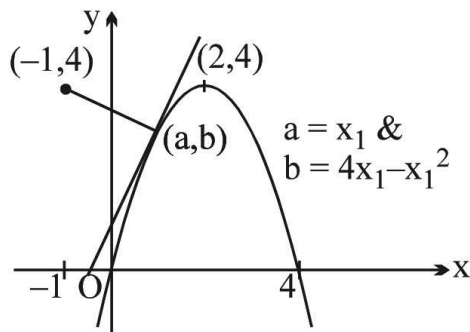
Ans. $2e^{-x/2}$

Sol. -

- Q13. Let $P(a, b)$ be a point on the parabola $y = 4x - x^2$ and is the point nearest to the point $A(-1, 4)$. Find $(a + b)$.

Ans. 4

Sol.



$$(2x_1 - 4)(4x_1 - x_1^2 - 4) = x_1 + 1$$

$$-2x_1^3 + 12x_1^2 - 25x_1 + 15 = 0$$

$$2x_1^3 - 12x_1^2 - 25x_1 + 15 = 0$$

$$2x_1^3 - 12x_1^2 - 25x_1 - 15 = 0$$

$$\therefore x_1 = 1; y_1 = 4 - 1 = 3$$

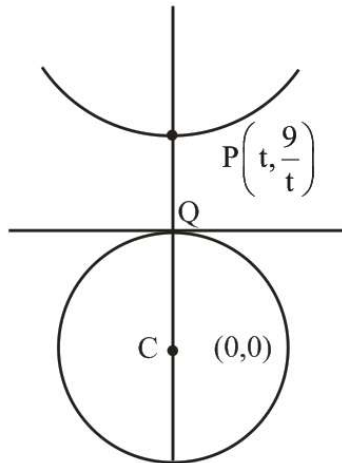
$$\therefore (a, b) = (1, 3) \Rightarrow a + b = 4$$

- Q14. Find the shortest distance between curves $xy = 9$ and $x^2 + y^2 = 1$

Ans. $3\sqrt{2} - 1$

Sol. $xy = 9$

$$x^2 + y^2 = 1$$



Minimum distance between curves will be along their common normal

Normal at P

$$y - \frac{9}{t} = mN(x - t)$$

$$1 \times y + x \frac{dy}{dx} = 0$$

$$y - \frac{9}{t} = \frac{t^2}{9}(x - t)$$

$$mT = \frac{-y}{x}$$

It passes through (0, 0)

$$mT = \frac{-9}{t^2}$$

$$0 - \frac{9}{t} = \frac{t^2}{9}(0 - t)$$

$$t^4 = 81$$

$$t = \pm 3$$

$$\text{if } t = 3$$

$$P(3, 3)$$

$$PQ = |cp - r|$$

$$PQ = |3\sqrt{2} - 1|$$

$$\text{if } t = -3$$

$$P(3, 3)$$

$$PQ = |cp - r|$$

$$= |3\sqrt{2} - 1|$$

Q15.

Ans. -

Sol. -

Q16.

Ans. -

Sol. -

Q17. Find the angle of intersection of the curves $y^2 = 4ax$ and $x^2 = 32ay$.

Ans. $\frac{\pi}{2}, \tan^{-1} \left(\frac{3}{5} \right)$

Sol. $y^2 = 4ax$ (i)

$x^2 = 32ay$ (ii)

Point of intersection of (i) & (ii)

$$x = \frac{y^2}{4a}$$

$$\frac{y^4}{16a^2} = 32ay$$

$$y^4 = 512a^3y^3$$

$$y = 0$$

$$y = 8a$$

$$x = 0$$

$$x = 16a$$

hence we have two point of intersection

(0, 0)

at origin tangent of $y^2 = 4ax$ is $x = 0$ and $x^2 = 32ay$ is $y = 0$

hence angle between tangent is $\frac{\pi}{2}$

(16a, 8a)

$$2y \frac{dy}{dx} = 4a$$

$$m_1 = \frac{2a}{y} = \frac{2a}{8a} = \frac{1}{4}$$

$$2x = 32a \frac{dy}{dx}$$

$$m_2 = \frac{x}{16a} = \frac{16a}{16a} = 1$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan\theta = \left| \frac{1 - \frac{1}{4}}{1 + \frac{1}{4}} \right|$$

$$\tan\theta = \frac{3}{5}$$

$$\theta = \tan^{-1}\left(\frac{3}{5}\right)$$

Q18. Show that the curves $\frac{x^2}{a^2 + K_1} + \frac{y^2}{b^2 + K_1} = 1$ & $\frac{x^2}{a^2 + K_2} + \frac{y^2}{b^2 + K_2} = 1$ intersect orthogonally.

Ans. -

Sol. -

Q19. If the two curves $C_1 : x = y^2$ and $C_2 : xy = k$ cut at right angles find the value of k.

Ans. $\pm \frac{1}{2\sqrt{2}}$

Sol. -

Q20. Find the value of n so that the subnormal at any point on the curve $xy^n = a^{n+1}$ may be constant.

Ans. (a)n=-2

Sol. Let P be any point on curve where $P\left(\frac{a^{n+1}}{\alpha^n}, \alpha\right)$

$$\text{Length of subnormal} = \left| y \cdot \frac{dy}{dx} \right|$$

$$1 \times y^n + nx y^{n-1} \frac{dy}{dx} = 0$$

$$y + nx \frac{dy}{dx} = 0$$

$$y + nx \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-y}{nx} = 0$$

$$\frac{dy}{dx} = \frac{-y}{nx}$$

$$\frac{dy}{dx} = \frac{-\alpha}{n(a^{n+1})}$$

$$(\alpha^3) = \frac{\alpha^{n+1}}{na^{n+1}}$$

$$\text{Length of subnormal} = \left| y \frac{dy}{dx} \right|$$

$$= \left| \frac{\alpha \cdot \alpha^{n+1}}{n \cdot a^{n+1}} \right|$$

$$= \left| \frac{\alpha \cdot \alpha^{n+2}}{n \cdot a^{n+1}} \right|$$

hence $n = -2$

Q21. Show that in the curve $y = a \ln(x^2 - a^2)$, sum of the length of tangent & subtangent varies as the product of the coordinates of the point of contact.

Ans. -

Sol. $y = a \ln(x^2 - a^2)$

Let co-ordinate of any point be (α, β)

$$\frac{dy}{dx} = \frac{\alpha a x}{x^2 - a^2}$$

$$\frac{dy}{dx} \text{ at } p = \frac{2a\alpha}{\alpha^2 - a^2}$$

Length of tangent + Length of subtangent

$$= \sqrt{1 + \left(\frac{dx}{dy}\right)^2} + y \left(\frac{dx}{dy}\right)$$

$$= B \sqrt{1 + \left(\frac{\alpha^2 - a^2}{2a\alpha}\right)^2}$$

$$= \frac{B(\alpha^2 + a^2)}{2a\alpha} + \frac{B(\alpha^2 - a^2)}{\alpha a \alpha}$$

$$= \frac{B}{2\alpha a}(\alpha^2)$$

$$= \frac{\alpha B}{\alpha a}$$

$$\text{hence sum} = \frac{\alpha B}{\alpha a}$$

$$\text{sum } \alpha 2B$$

Hence proved.

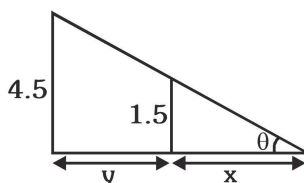
Q22. A man 1.5 m tall walks away from a lamp post 4.5 m high at the rate of 4 km/hr.

(i) how fast is the farther end of the shadow moving on the pavement ?

(ii) how fast is his shadow lengthening ?

Ans. (i) 6 km/h; (ii) 2 km/hr

Sol.



$$\frac{4.5}{x + y} = \frac{1.5}{x} \Rightarrow 3x = x + y \Rightarrow 2x = y$$

$$\frac{2dx}{dt} = \frac{dy}{dt}$$

$$\frac{dy}{dx} = 4 \text{ is given then } \frac{dx}{dt} = 2$$

$$(a) \frac{d}{dt}(x + y) = \frac{dx}{dt} + \frac{dy}{dt} = 4 + 2 = 6 \text{ km/hr.}$$

$$(b) \text{ Shadow lengthening} = \frac{dx}{dt} = 2 \text{ km/hr.}$$

Q23. A particle moves along the curve $6y = x^3 + 2$. Find the points on the curve at which the y coordinate is changing 8 times as fast as the x coordinate.

Ans. $(4, 11)$ & $(-4, -31/3)$

Sol. given

$$8 \frac{dx}{dt} = \frac{dy}{dt}$$

$$6 \frac{dy}{dt} = 3x^2 \frac{dx}{dt}$$

$$6 \times 8 \frac{dx}{dt} = 3x^2 \frac{dx}{dt}$$

$$x^2 = 16$$

$$x = \pm 4$$

$$y = \frac{(\pm 4)^3 + 2}{6}$$

$$y = \frac{\pm 64 + 2}{6}$$

$$y = 11 \text{ \& } -\frac{31}{3}$$

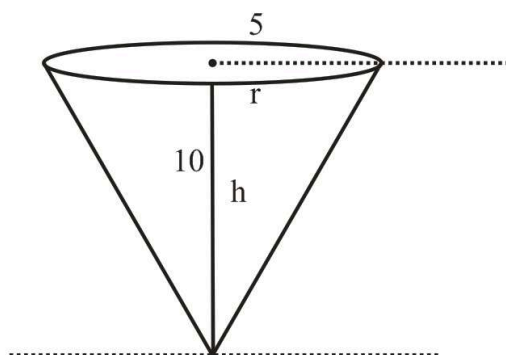
hence points are

$$(4, 11) \text{ \& } \left(-4, -\frac{31}{3}\right)$$

Q24. An inverted cone has a depth of 10 cm & a base of radius 5 cm. Water is poured into it at the rate of $1.5 \text{ cm}^3/\text{min}$. Find the rate at which level of water in the cone is rising, when the depth of water is 4 cm.

Ans. $3/8\pi \text{ cm/min}$

Sol.



$$V = \frac{1}{3}\pi r_1 h$$

$$\frac{dv}{dt} = 1.5 \text{ cm/min}$$

$$\frac{r}{h} = \frac{5}{10}$$

$$r = \frac{h}{2}$$

$$V = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h$$

$$V = \frac{\pi h^3}{12}$$

$$\frac{dv}{dt} = \frac{3\pi h^2}{12} \quad \frac{dh}{dt}$$

$$\frac{15}{10} = \frac{3\pi 16}{12} \quad \frac{dh}{dt}$$

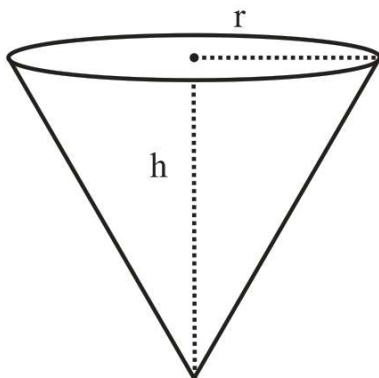
$$\frac{15}{10} = \frac{3\pi 16}{12} \quad \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{3}{8\pi} \text{ cm/min}$$

- Q25. A water tank has the shape of a right circular cone with its vertex down. Its altitude is 10 cm and the radius of the base is 15 cm. Water leaks out of the bottom at a constant rate of 1 cu. cm/sec. Water is poured into the tank at a constant rate of C cu. cm/sec. Compute C so that the water level will be rising at the rate of 4 cm/sec at the instant when the water is 2 cm deep.

Ans. $1+36\pi$ cu.cm/sec

Sol.



$$\frac{r}{h} = \frac{15}{10}$$

$$r = \frac{3}{2}h$$

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi\left(\frac{9h^3}{4}\right)$$

$$V = \frac{9\pi h^3}{12}$$

$$\frac{dv}{dt} = \frac{27\pi h^2}{12} \quad \frac{dh}{dt} = C - 1$$

$$\frac{dv}{dt} = \frac{27\pi h^2}{12}(4)(4) = C - 1$$

$$C = 1 + 36\pi$$

Q26. Sand is pouring from a pipe at the rate of 12 cc/sec. The falling sand forms a cone on the ground in such a way that the height of the cone is always 1/6th of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm.

Ans. $\frac{1}{48} \pi$ cm/s

Sol. -

Q27.

Ans. -

Sol. -

Q28. Find the intervals of monotonicity for the following functions & represent your solution set on the number line.

$$(a) f(x) = 2e^{x^2} - 4x \quad (b) f(x) = \frac{e^x}{x} \quad (c) f(x) = x^2 e^{-x} \quad (d) f(x) = 2x^2 - \ln |x|$$

Also plot the graphs in each case & state their range.

Ans. (a) I in $(2, \infty)$ & D in $(-\infty, 2)$

(b) I in $(1, \infty)$ & D in $(-\infty, 0) \cup (0, 1)$

(c) I in $(0, 2)$ & D in $(-\infty, 0) \cup (2, \infty)$

(d) I for $x > \frac{1}{2}$ or $-\frac{1}{2} < x < 0$ & D for $x < -\frac{1}{2}$ or $0 < x < \frac{1}{2}$

Sol. (a) $f(x) = 2e^{x^2-4x}$

$$f'(x) = 2e^{x^2-4x}(2x-4)$$

If function is increasing $f'(x) > 0$

hence $x > 2$

If function is decreasing

$$f'(x) < 0$$

hence $x < 2$

$$(b) f(x) = \frac{e^x}{x}$$

$$f'(x) = \frac{e^x - e^x}{x^2}$$

$$f'(x) = \frac{e^x(x-1)}{x^2}$$

If function is increasing

$$f'(x) > 0$$

$$x > 1$$

If function is decreasing

$$f'(x) < 0$$

$$x < 1$$

hence $x \in (-\infty, 1) - \{0\}$

$$(c) f'(x) = x^2 e^{-x}$$

$$f'(x) = x^2 x e^{-x} - x^2 e^{-x}$$

$$f'(x) = e^{-x}(2x - x^2)$$

$$f'(x) = e^{-x}(2x - x)(x)$$

for increasing $f'(x) > 0$

$$x \in (0, 2)$$

for decreasing $f'(x) < 0$

$$x \in (-\infty, 0) \cup (2, \infty)$$

$$(D) f(x) = 2x^2 - \ln|x|$$

$$f(x) = \begin{cases} 2x^2 - \ln x & ; \quad x > 0 \\ 2x^2 - \ln(-x) & ; \quad x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 4x - \frac{1}{x} & ; \quad x > 0 \\ 4x - \frac{1}{x} & ; \quad x < 0 \end{cases}$$

$$f'(x) = \begin{cases} \frac{4x^2 - 1}{x} & ; \quad x > 0 \\ \frac{4x^2 - 1}{x} & ; \quad x < 0 \end{cases}$$

For increasing $f'(x) > 0$

$$\frac{4x^2 - 1}{x} > 0$$

$$\frac{(2x+1)(2x-1)}{x} > 0$$

$$x \in \left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$$

For decreasing $f'(x) < 0$

$$x \in (-\infty, -\frac{1}{2}) \cup (0, \frac{1}{2})$$

Q29. Find the intervals of monotonicity of the functions in $[0, 2\pi]$

(a) $f(x) = \sin x - \cos x$ in $x \in [0, 2\pi]$ (b) $g(x) = 2 \sin x + \cos 2x$ in $(0 \leq x \leq 2\pi)$.

(c) $f(x) = \frac{4 \sin x - 2x - x \cos x}{2 + \cos x}$

Ans. (a) I in $[0, 3\pi/4) \cup (7\pi/4, 2\pi]$ & D in $(3\pi/4, 7\pi/4)$

(b) I in $[0, \pi/6) \cup (\pi/2, 5\pi/6) \cup (3\pi/2, 2\pi]$ & D in $(\pi/6, \pi/2) \cup (5\pi/6, 3\pi/2)$

(c) I in $[0, \pi/2) \cup (3\pi/2, 2\pi]$ and D in $(\pi/2, 3\pi/2)$

Sol. -

Q30. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x) = \begin{cases} \max\{f(t) : 0 \leq t \leq x\} & , 0 \leq x \leq 1 \\ 3 - x & , 1 < x \leq 2 \end{cases}$

Discuss the conti. & differentiability of $g(x)$ in the interval $(0, 2)$.

Ans. continuous but not diff. at $x = 1$

Sol. -

Q31. Let $f(x) = 1 - x - x^3$. Find all real values of x satisfying the inequality, $1 - f(x) - f^3(x) > f(1 - 5x)$

Ans. $(-2, 0) \cup (2, \infty)$

Sol. -

Q32. Let $f(x)$ be an increasing function defined on $(0, \infty)$. If $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$. Find the range of a .

Ans. $(0, 1/3) \cup (1, 5)$

Sol. -

Q33. Find the set of all values of the parameter 'a' for which the function,

$f(x) = \sin 2x - 8(a + 1)\sin x + (4a^2 + 8a - 14)x$ increases for all $x \in \mathbb{R}$ and has no critical points for all $x \in \mathbb{R}$.

Ans. $a < -(2 + \sqrt{5})$ or $a > \sqrt{5}$

Sol. $f(x) = \sin 2x - 8(a + 1)\sin x + (4a^2 + 8a - 14)x$

$$f'(x) = 2 \cos 2x - 8(a + 1) \cos x + (4a^2 + 8a - 14)$$

$$f'(x) = 2(2\cos^2 x - 1) - 8(a + 1) \cos x + 4a^2 + 8a - 14$$

$$= 4\{\cos^2 x - 2(a + 1) \cos x\} + 4a^2 + 8a - 16$$

$$= 4\{\cos x - (a + 1)\}^2 - 20 > 0$$

$$= \{\cos x - (a + 1)\}^2 - (\sqrt{5})^2 > 0$$

$$f'(x) = \{\cos x - (a + 1)\} \{\cos x - \sqrt{5}(a + 1) + \sqrt{5}\} > 0$$

$$\Rightarrow \cos x > a + 1 + \sqrt{5} \text{ or } \cos x < (a + 1) - \sqrt{5} \forall x \in \mathbb{R}$$

$$a + 1 + \sqrt{5} < -1 \text{ or } (a + 1) - \sqrt{5} > 1$$

$$a < -2 - \sqrt{5} \text{ or } a > \sqrt{5}$$

$$a \in (-\infty, -2 - \sqrt{5}) \cup (\sqrt{5}, \infty)$$

Q34. Find all the values of the parameter 'a' for which the function ;

$f(x) = 8ax - a \sin 6x - 7x - \sin 5x$ increases & has no critical points for all $x \in \mathbb{R}$.

Ans. $(6, \infty)$

Sol. $f(x) = 8ax - a \sin 6x - 7x - \sin 5x$

$$f'(x) = 8a - 6a \cos 6x - 7 - 5 \cos 5x > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow a > \frac{7 + 5 \cos 5x}{8 - 6 \cos 6x} \quad \forall x \in \mathbb{R}$$

$\Rightarrow$ RHS assumes maximum value for $x = 0$.

$$\Rightarrow a > \frac{7 + 5}{8 - 6} \Rightarrow a > 6$$

Q35.

Ans. -

Sol. -

Q36. Find the values of 'a' for which the function $f(x) = \sin x - a \sin 2x - \frac{1}{3} \sin 3x + 2ax$ increases throughout the number line.

Ans. $[1, \infty)$

Sol. $f'(x) = \cos x - 2a \cos 2x - \cos 3x + 2a$

for always increasing

$$f'(x) > 0$$

$$\cos x - \cos 3x + 2a(1 - \cos 2x) \geq 0$$

$$2a(1 - \cos 2x) \leq \cos 3x - \cos x$$

$$2a(2 \sin^2 x) \geq -2 \sin 2x$$

$$a \leq -\cos x$$

hence $a \geq 1$

Q37. Find the set of values of 'a' for which the function,

$f(x) = \left(1 - \frac{21 - 4a - a^2}{a + 1}\right)x^3 + 5x + \sqrt{7}$ is increasing at every point of its domain.

Ans. $[-7, -1) \cup [2, 3]$

Sol. -

Q38. If $f(x) = 2e^x - ae^{-x} + (2a + 1)x - 3$ monotonically increases for every $x \in \mathbb{R}$ then find the range of values of 'a'.

Ans. $a \geq 0$

Sol. $f(x) = 2e^x - ae^{-x} + (2a + 1)x - 3$

$$f'(x) = 2e^x + ae^{-x} + (2a + 1)$$

Let $e^x = t$; $t > 0$

$$\therefore 2t + \frac{a}{t}(2a + 1) \Rightarrow \frac{2t^2 + (2a + 1)t + a}{t}$$

$$\Rightarrow \text{Now } f'(x) \geq 0 \quad \therefore 2t^2 + (2a + 1)t + a \geq 0$$

Case-I

$$(2a + 1)^2 - 8a \leq 0$$

$$\Rightarrow (2a - 1)^2 \leq 0 \Rightarrow a = \frac{1}{2}$$

Case-II



$$\Rightarrow \text{(i) } D \geq 0 \Rightarrow a \in \mathbb{R}$$

$$\text{(ii) } f(0) \geq 0 \Rightarrow a \geq 0$$

$$\text{(iii) } -\frac{b}{2a} < 0 \Rightarrow \frac{-2(a + 1)}{4} < 0 \Rightarrow a > -\frac{1}{2}$$

Q39. Find the set of values of x for which the inequality $\ln(1 + x) > x/(1 + x)$ is valid.

Ans. $(-1, 0) \cup (0, \infty)$

Sol. -

Q40. Prove that, $x^2 - 1 > 2x \ln x > 4(x - 1) - 2 \ln x$ for $x > 1$.

Ans. .

Sol. $x^2 - 1 > 2x \ln x > 4(x - 1) - 2 \ln x, x > 1$

(a) Consider $f(x) = x^2 - 1 - 2x \ln x$

$$f'(x) = 2\{x - 1 - \ln x\}$$

$$f''(x) = 2\left\{1 - \frac{1}{x}\right\}$$

$$f''(x) > 0 \quad \forall x > 1$$

$$\Rightarrow f(x) \text{ is increasing } \forall x > 1$$

$$\Rightarrow f'(x) > f'(1) \Rightarrow f'(x) > 0 \quad \forall x > 1$$

$$\Rightarrow f(x) \text{ is increasing } \forall x > 1$$

$$\Rightarrow f(x) > f(1) \Rightarrow f(x) > 0 \quad \forall x > 1$$

(b) Consider,

$$g(x) = 2x \ln x + 2 \ln x - 4(x - 1)$$

$$g'(x) = 2(1 + \ln x) + \frac{2}{x} - 4$$

$$g''(x) = \frac{2}{x} - \frac{2}{x^2}$$

$$g''(x) = \frac{2(x-1)}{x^2} > 0 \forall x > 1$$

$$\Rightarrow g'(x) \text{ is increasing } \forall x > 1$$

$$\Rightarrow g'(x) > g'(1) \Rightarrow g'(x) > 0 \forall x > 1$$

$$\Rightarrow g(x) \text{ is increasing } \forall x > 1$$

$$\Rightarrow g(x) > g(1) \Rightarrow x > 1$$

$$\Rightarrow g(x) > 0 \forall x > 1.$$

Q41. Let f, g be differentiable on $\mathbb{R}$ and suppose that $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$. Show that $f(x) \leq g(x)$ for all $x \geq 0$.

Ans. -

Sol. We have to prove

$$f(x) \leq g(x)$$

$$f(x) - g(x) \leq 0$$

$$\text{Let } h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$\text{but } f'(x) \leq g'(x)$$

$$\text{hence } h'(x) < 0$$

$$h(x) \text{ is decreasing}$$

$$\text{So } h(x) \leq h(0)$$

$$f(x) - g(x) \leq 0$$

$$f(x) \leq g(x)$$

Q42. Show that exactly two real values of x satisfy the equation $x^2 = x \sin x + \cos x$.

Ans. -

$$\text{Sol. } x^2 = x \sin x + \cos x$$

$$x^2 - x \sin x - \cos x = 0$$

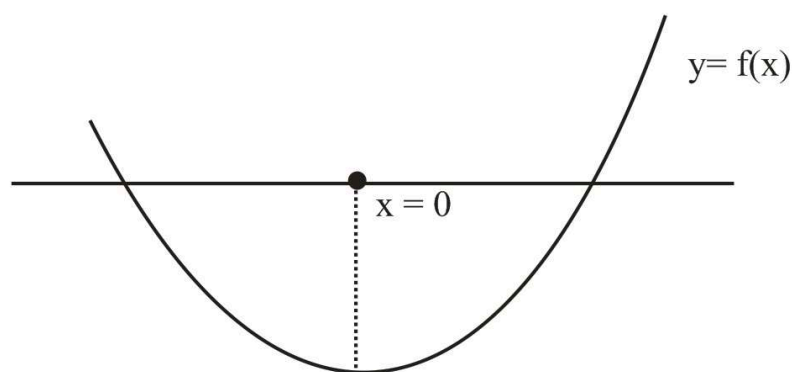
$$f(x) = x^2 - x \sin x - \cos x$$

$$f'(x) = 2x - \sin x - x \cos x + \sin x$$

$$f'(x) = 2x - x \cos x$$

$$f'(x) = x(2 - \cos x)$$

graph



hence it is clerr there will be exactly two solution.

Q43. Find the greatest & the least values of the following functions in the given interval if they exist.

(a) $f(x) = \sin^{-1} \frac{x}{\sqrt{x^2+1}} - \ln x$ in $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ (b) $f(x) = 12x^{4/3} - 6x^{1/3}$, $x \in [-1, 1]$

(c) $y = x^5 - 5x^4 + 5x^3 + 1$ in $[-1, 2]$

Ans. (a) $(\pi/6) + (1/2)\ln 3$, $(\pi/3) - (1/2)\ln 3$

(b) Maximum at $x = 1$ and $f(-1) = 18$; Minimum at $x = 1/8$ and $f(1/8) = -9/4$

(c) 2 & -10

Sol. -

Q44. Find the minimum value of the function $f(x) = 4^{\frac{3}{2}} + x^{\frac{-3}{2}} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x .

Ans. -10

Sol. -

Q45. If $b > a$, find the minimum value of $|(x-a)^3| + |(x-b)^3|$, $x \in \mathbb{R}$.

Ans. $(b-a)^{3/4}$

Sol. -

Q46. Let $f(x) = 4x^3 - 3x^2 - 2x + 1$, use Rolle's theorem to prove that there exist c , $0 < c < 1$ such that $f(c) = 0$.

Ans. -

Sol. Let

$$h'(x) = 4x^3 - 3x^2x + 1$$

$$h'(x) = x^4 - x^3 - x^2 + x + c$$

Rolle's theorm in $[0, 1]$

(1) Continuous in $[0, 1]$

(2) differentiable in $(0, 0)$

(3) $h(0) = h(1)$

hence in interval $(0, 1)$, there exists at least a value where

$$h'(x) = 0$$

$$\text{or } 4x^3 - 3x^2 - 2x + 1 = 0 \text{ in } (0, 1).$$

Q47. Let f and g be functions, continuous in $[a, b]$ and differentiable on $[a, b]$. If $f(a) = f(b) = 0$ then show that there is a point $c \in (a, b)$ such that $g'(c)f(c) + f'(c)g(c) = 0$.

Ans. -

Sol. -

Q48. Assume that f is continuous on $[a, b]$, $a > 0$ and differentiable on an open interval (a, b) .

Show that if $\frac{f(a)}{a} = \frac{f(b)}{b}$, then there exist $x_0 \in (a, b)$ such that $x_0 f'(x_0) = f(x_0)$.

Ans. -

Sol. -

Q49. $f(x)$ and $g(x)$ are differentiable functions for $0 \leq x \leq 2$ such that $f(0) = 5$, $g(0) = 0$, $f(2) = 8$, $g(2) = 1$. Show that there exists a number c satisfying $0 < c < 2$ and $f'(c) = 3g'(c)$.

Ans. -

Sol. $h(x) = f(x) - 3g(x)$

Clearly $h(x)$ is continuous and differentiable in $[0, 2]$ and $h(0) = h(2)$

hence in $[0, 2]$ Rolle's theorem can be applied.

$$h'(x) = 0 \text{ in } [0, 2]$$

$$f'(x) - 3g'(x) = 0$$

$$f'(x) - 3g'(x)$$

Q50. If $p(x) = 51x^{101} - 2323x^{100} - 45x + 1035$, using Rolle's theorem, prove that at least one root of $p(x)$ lies between $(45^{1/100}, 46)$.

Ans. -

Sol. -

Q51. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(a) = 0$, $f(b) = 2$, $f(c) = -1$, $f(d) = 2$, $f(e) = 0$, where $a < b < c < d < e$. Determine the minimum number of zeroes of $g(x) = (f(x))^2 + f(x) \cdot f'(x)$ in the interval $[a, e]$.

Ans. 6

Sol. -

Q52. Let f be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = a$ and $f(b) = b$, show that there exist distinct c_1, c_2 in (a, b) such that $f'(c_1) + f'(c_2) = 2$.

Ans. -

Sol. By LMVT in $\left[a, \frac{a+b}{2}\right]$

$$\frac{f\left(\frac{a+b}{2}\right) - f(a)}{\frac{a+b}{2} - a} = f'(c_1), c_1 \in \left(a, \frac{a+b}{2}\right) \quad \dots\dots(i)$$

By LMVT in $\left[\frac{a+b}{2}, b\right]$

$$\frac{f(b) - f\left(\frac{a+b}{2}\right)}{b - \frac{a+b}{2}} = f'(c_2), c_2 \in \left(\frac{a+b}{2}, b\right)$$

$$(i) + (ii) \Rightarrow f'(c_1) + f'(c_2) = 2$$

Q53. For what value of a, m and b does the function $f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \leq x \leq 2 \end{cases}$ satisfy the hypothesis of the mean value theorem for the interval $[0, 2]$.

Ans. $a = 3, b = 4$ and $m = 1$

Sol. -

Q54. Let $a > 0$ and f be continuous in $[-a, a]$. Suppose that $f'(x)$ exists and $f'(x) \leq 1$ for all $x \in (-a, a)$. If $f(a) = a$ and $f(-a) = -a$, show that $f(0) = 0$.

Ans. -

Sol. -

Q55. Let f be a differentiable function for all x and that $|f'(x)| \leq 2$ for all x . If $f(1) = 2$ and $f(4) = 8$ then compute the value of $f^2(2) + f^2(3)$.

Ans. 52

Sol. -

Q56. Find the greatest & least value for the function ;

$$(a) y = x + \sin 2x, 0 \leq x \leq 2\pi \quad (b) y = 2 \cos 2x - \cos 4x, 0 \leq x \leq \pi$$

Ans. (a) Max at $x=2\pi$, Max value $=2\pi$, Min. at $x=0$, Min value $=0$

(b) Max at $x=\pi/6$ & also at $x=5\pi/6$ and Max value $=3/2$, Min at $x=\pi/2$, Min value $=-3$

Sol. (a) $y = x + \sin 2x : 0 \leq x \leq 2\pi$

$$\frac{dy}{dx} = 1 + 2\cos 2x = 0$$

$$\cos 2x = \frac{-1}{2}$$

$$2x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

critical points

$$x = \frac{\pi}{3}, \frac{2\pi}{3}$$

$$\text{Greatest value} = \text{Max} \left\{ f(0), f\left(\frac{\pi}{3}\right), f\left(\frac{2\pi}{3}\right), f(2\pi) \right\}$$

$$= \text{Max} \left(0, \frac{\pi}{3} + \frac{\sqrt{3}}{2}, \frac{2\pi}{3} - \frac{\sqrt{3}}{2}, 2\pi \right)$$

$$= \text{Max } 2\pi$$

Minimum value $= 0$

$$y = 2\cos 2x - \cos 4x : 0 \leq x \leq \pi$$

$$\frac{dy}{dx} = -4\sin 2x + 4\sin 4x = 0$$

$$\sin 4x - \sin 2x = 0$$

$$2\cos 3x \sin x = 0$$

$$\cos 3x = 0$$

$$\sin x = 0$$

$$3x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

$$x = r\pi$$

$$x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$$

$$x = 0, \pi$$

$$\text{Greatest value} = \text{Max} \left(f(0), f\left(\frac{\pi}{6}\right), f(\pi), f\left(\frac{\pi}{2}\right), f\left(\frac{5\pi}{6}\right) \right)$$

$$= \text{Max} \left(1, \frac{3}{2}, 1, -3, \frac{3}{2} \right)$$

$$= \frac{3}{2}$$

$$\text{Minimum value} = 1 - 3 \text{ at } x = \frac{\pi}{2}$$

Q57. Let $P(x)$ be a polynomial of degree 5 having extremum at $x = -1, 1$ and

$$\lim_{x \rightarrow 0} \left(\frac{P(x)}{x^3} - 2 \right) = 4.$$

If M and m are the maximum and minimum value of the function $y = P'(x)$ on the set $A = \{x | x^2 + 6 \leq 5x\}$ then find $\frac{m}{M}$.

Ans. 6

Sol. -

Q58. Of all the lines tangent to the graph of the curve $y = \frac{6}{x^2 + 3}$, find the equations of the tangent lines of minimum and maximum slope.

Ans. $3x + 4y - 9 = 0$; $3x - 4y + 9 = 0$

Sol. -

Q59. If $y = \frac{ax + b}{(x - 1)(x - 4)}$ has a turning value at $(2, -1)$ find a & b and show that the turning value is a maximum.

Ans. $a = 1, b = 0$

Sol. -

Q60. Let $f(x)$ be a cubic polynomial which has local maximum at $x = -1$ and $f'(x)$ has a local minimum at $x = 1$. If $f(-1) = 10$ and $f(3) = -22$, then find the distance between its two horizontal tangents.

Ans. 32

Sol. -

Q61. Consider the function $f(x) = \begin{cases} \sqrt{x} \ln x & \text{when } x > 0 \\ 2 & \text{for } x = 0 \end{cases}$

(a) Find whether f is continuous at $x = 0$ or not.

(b) Find the minima and maxima if they exist.

(c) Does $f'(0)$ exist? Find $\lim_{x \rightarrow 0} f'(x)$.

(d) Find the inflection points of the graph of $y = f(x)$.

Ans. (a) f is continuous at $x = 0$; (b) $-2/e$; (c) does not exist, does not exist ; (d) pt. of inflection $x = 1$

Sol. -

Q62. If $P(x)$ be a polynomial of degree 3 satisfying $P(-1) = 10$, $P(1) = -6$ and $P(x)$ has maximum at $x = -1$ and $P'(x)$ has minima at $x = 1$. Find the distance between the local maximum and local minimum of the curve.

Ans. $4\sqrt{65}$

Sol. Let $P(x) = ax^3 + bx^2 + cx + d$
 $P'(x) = 3ax^2 + 2bx + c$
 $P''(x) = 6ax + 2b$
given that $P(-1) = -a + b - c + d = 10$,
 $P(1) = a + b + c + d = -6$
 $P'(-1) = 3a - 2b + c = 0$
 $P''(1) = 6a + 2b = 0$
 $\therefore a = 1, b = -3, c = -9, d = 5$
 $\Rightarrow P(x) = x^3 - 3x^2 - 9x + 5$
 $\Rightarrow P'(x) = 3x^2 - 6x - 9 = 3(x + 1)(x - 3)$
 $x = -1$ is point of maxima & $x = 3$ is point of minima
Hence distances between $(-1, 10)$ and $(3, -22)$ is $4\sqrt{65}$ units.

Q63. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set
 $A = \{x | x^2 + 20 \leq 20 \leq 9x\}$ is

Ans. 7

Sol. $f(x) = 2x^3 - 15x^2 + 36x - 48$
Set $A = \{x | x^2 + 20 \leq 9x\}$
 $\therefore x^2 - 9x + 20 \leq 0$
 $(x - 5)(x - 4) \leq 0$
 $\Rightarrow x \in [4, 5]$
Now, $f'(x) = 6x^2 - 30x + 36 = 0$
 $\Rightarrow x^2 - 5x + 6 = 0$
 $x = 2, 3$ and $f(x) \uparrow$ in $x \in (-\infty, 2) \cup (3, \infty)$
 $\Rightarrow$ In the set A , $f(x)$ is increasing
 $f(x)_{\max} = f(5)$
 $= 2.125 - 15.25 + 36.5 - 48$
 $= 7$

Q64. Let f be a function defined on $\mathbb{R}$ (the set of all real numbers) such that $f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$, for all $x \in \mathbb{R}$.

If g is a function defined on $\mathbb{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, for all $x \in \mathbb{R}$, then the number of points in $\mathbb{R}$ at which g has a local maximum is

Ans. 1

Sol. $f(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4 \forall x \in \mathbb{R}$
 $f(x) = \ln(g(x)) \forall x \in \mathbb{R}$
 $g(x) = e^{f(x)}$
 $g'(x) = 0 \Rightarrow e^{f(x)} \cdot f'(x) = 0 \Rightarrow f'(x) = 0$

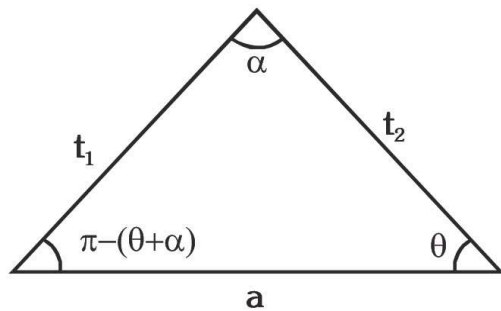
increasing	decreasing	decreasing	increasing	increasing				
+	2009	-	2010	-	2011	+	2012	+

local maximum at $x = 2009$, hence only 1 point.

Q65. Find the maximum perimeter of a triangle on a given base 'a' and having the given vertical angle α .

Ans. $P_{max} = a \left(1 + \operatorname{cosec} \frac{\alpha}{2} \right)$

Sol.



$$\frac{a}{\sin \alpha} = \frac{t_1}{\sin \theta} = \frac{t_2}{\sin(\theta + \alpha)}$$

$$t_1 = \frac{a}{\sin \theta} \sin \theta, \quad t_2 = \frac{a}{\sin \alpha} \sin(\theta + \alpha)$$

to maximize perimeter we maximize $t_1 + t_2$

$$t_1 + t_2 = f(\theta) = 2 \frac{a}{\sin \alpha} \{ \sin(\theta) + \sin(\theta + \alpha) \}$$

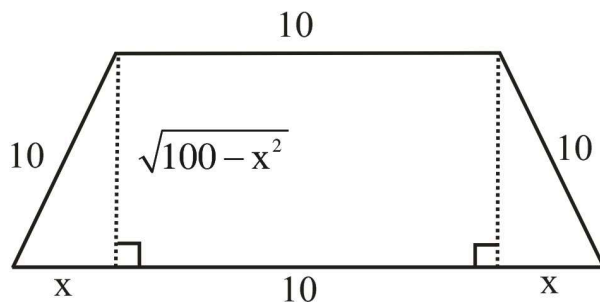
$$= \frac{a}{\sin \alpha} \left\{ \sin \left(\theta + \frac{\alpha}{2} \right) \cos \frac{\alpha}{2} \right\}$$

$$\text{for maximum } \sin \left(\theta + \frac{\alpha}{2} \right) = 1$$

Q66. The length of three sides of a trapezium are equal, each being 10 cms. Find the maximum area of such a trapezium.

Ans. $75\sqrt{3}$ sq.units

Sol.



$$A = \frac{1}{2}(2x + 10 + 10)\sqrt{100 - x^2}$$

$$A = (x + 10)\sqrt{100 - x^2}$$

$$\begin{aligned}\frac{dA}{dx} &= \sqrt{100 - x^2} + (x + 10) \frac{1(-2x)}{2\sqrt{100 - x^2}} \\ &= \frac{(100 - x^2) + (x + 10)(-x)}{\sqrt{100 - x^2}} \\ &= \frac{-2(x + 10)(x - 5)}{\sqrt{100 - x^2}}\end{aligned}$$

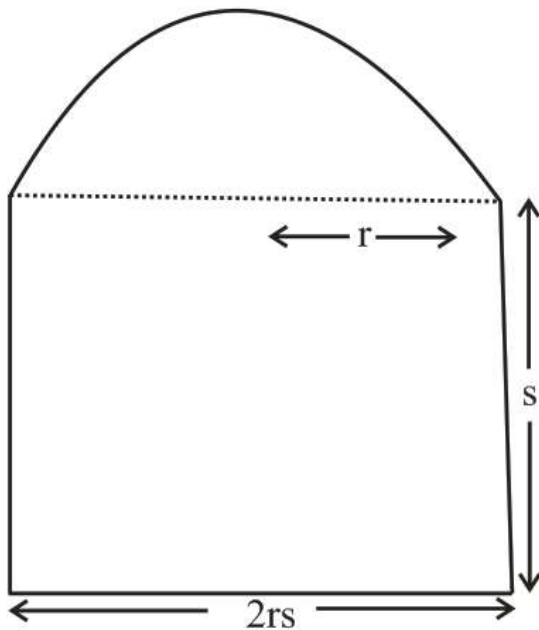
Clearly $x = 5$ gives maximum area

$$\begin{aligned}A &= (5 + 10)\sqrt{100 - 25} \\ &= 15\sqrt{75} \\ &= 75\sqrt{3}\end{aligned}$$

Q67. The plan view of a swimming pool consists of a semicircle of radius r attached to a rectangle of length ' $2r$ ' and width ' s '. If the surface area A of the pool is fixed, for what value of ' r ' and ' s ' the perimeter ' P ' of the pool is minimum.

Ans. $r = \sqrt{\frac{2A}{\pi + 1}}, s = \sqrt{\frac{2A}{\pi + 4}}$

Sol.



$$A = 2rs + \frac{\pi r^2}{2} \quad (\text{Fixed}) \quad \dots\dots (i)$$

$$p = 2r + 2s + \pi r$$

$$p = 2r + \pi r + 2\left(A - \frac{\pi r^2}{2}\right) \times \frac{1}{2r}$$

$$p = 2r + \frac{A}{\pi r} + \frac{A}{r} - \frac{\pi r}{2}$$

$$p = 2r + \frac{A}{\pi r} + \frac{\pi r}{2}$$

$$\frac{dp}{dr} = 2 - \frac{A}{r^2} + \frac{\pi}{2} = 0$$

$$2 + \frac{\pi}{2} = \frac{A}{r^2}$$

$$\frac{4 + \pi}{2} = \frac{A}{r^2}$$

$$r = \sqrt{\frac{2A}{4 + \pi}}$$

gives 'p' maximum

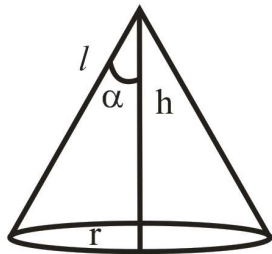
by (i) we get

$$s = \sqrt{\frac{2A}{4 + \pi}}$$

Q68. For a given curved surface of a right circular cone when the volume is maximum, prove that the semi vertical angle is $\sin^{-1} \frac{1}{\sqrt{3}}$.

Ans. -

Sol.



$$A = \pi r l \text{ (given)}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi r^2 \sqrt{l^2 - r^2}$$

$$V = \frac{1}{3} \pi r^2 \sqrt{\frac{A^2}{\pi^2 r^2} - r^2}$$

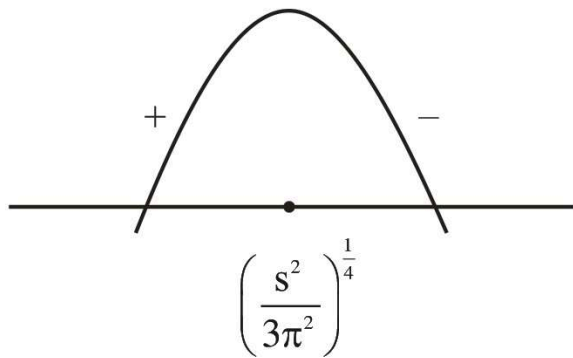
$$V = \frac{1}{3} \sqrt{\frac{A^2 - \pi^2 r^4}{\pi^2}}$$

$$V = \frac{1}{3} r \sqrt{A^2 - \pi^2 r^4}$$

$$\frac{dv}{dr} = \frac{1}{3} \sqrt{A^2 - \pi^2 r^4} + \frac{1}{3} r \left(\frac{-4\pi^2 r^3}{2\sqrt{A^2 - \pi^2 r^4}} \right)$$

$$= \frac{1}{3} \left(\frac{A^2 - \pi^2 r^4 - 2\pi^2 r^4}{\sqrt{A^2 - \pi^2 r^4}} \right)$$

$$\frac{dv}{dr} = \frac{1}{3} \frac{(A^2 - 3\pi^2 r^4)}{\sqrt{A^2 - \pi^2 r^4}}$$



for $\frac{dv}{dr} = 0$

$$r = \left(\frac{A^2}{3\pi^2}\right)^{\frac{1}{4}}$$

hence $V_{\max}$ is at $r = \left(\frac{A^2}{3\pi^2}\right)^{\frac{1}{4}}$

$$\sin\alpha = \frac{r}{l} = \frac{r}{\frac{A}{\pi r}} = \frac{\pi r^2}{A}$$

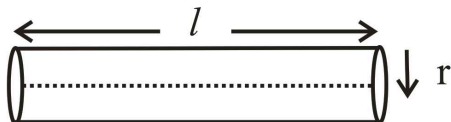
$$= \frac{\pi}{A} \left(\frac{A^2}{3\pi^2}\right)^{\frac{1}{2}} = \frac{1}{\sqrt{3}}$$

$$\alpha = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Q69. By the post office regulations, the combined length & girth of a parcel must not exceed 3 metre. Find the volume of the biggest cylindrical (right circular) packet that can be sent by the parcel post.

Ans. $1/\pi$ cum

Sol.



$$l + 2\pi r = 3$$

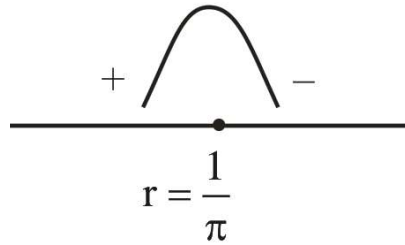
$$V = \pi r^2 l$$

$$V = \pi r^2 (3 - 2\pi r)$$

$$V = 3\pi r^2 - 2\pi^2 r^3$$

$$\frac{dv}{dr} = 6\pi r - 6\pi^2 r^2$$

$$= 6\pi r(1 - \pi r)$$



$$= 6\pi^2 r \left(\frac{1}{\pi} - r \right)$$

hence maximum volume will be 3 at $r = \frac{1}{\pi}$

$$l + 2\pi r = 3$$

$$l + 2\pi \left(\frac{1}{\pi} \right) = 3$$

$$l = 1$$

$$V_{max} = \pi r^2 l$$

$$= \pi \left(\frac{1}{\pi^2} \right) (1)$$

$$V_{max} = \frac{1}{\pi}$$

- Q70. A running track of 440 ft. is to be laid out enclosing a football field, the shape of which is a rectangle with semi circle at each end. If the area of the rectangular portion is to be maximum, find the length of its sides.

Use : $\pi \approx 22/7$.

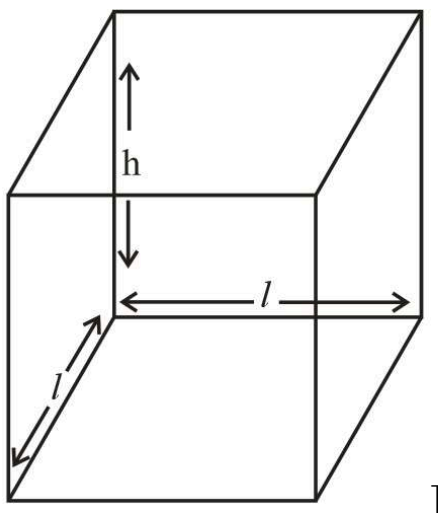
Ans. 110 ' , 70 '

Sol. -

- Q71. A closed rectangular box with a square base is to be made to contain 1000 cubic feet. The cost of the material per square foot for the bottom is 15 paise, for the top 25 paise and for the sides 20 paise. The labour charges for making the box are Rs. 3/-. Find the dimensions of the box when the cost is minimum.

Ans. side 10', height 10'

Sol.



$$l^2 h = 1000$$

$C = \text{cost of bottom} + \text{Cost of top} + \text{Cost of sides}$

$$= l^2 \times 150 + l^2 \times 25 + 4(lh)20$$

$$C = 40l^2 + 80lh$$

$$C = 40l^2 + 80l \left(= \frac{1000}{l^2} \right)$$

$$C = 40l^2 + \frac{80000}{l}$$

$$C = 40 \left(l^2 + \frac{2000}{l} \right)$$

$$\frac{dc}{dl} = 40 \left(2l - \frac{2000}{l^2} \right)$$

$$= \frac{80(l^3 - 1000)}{l^2}$$

hence $l = 10$

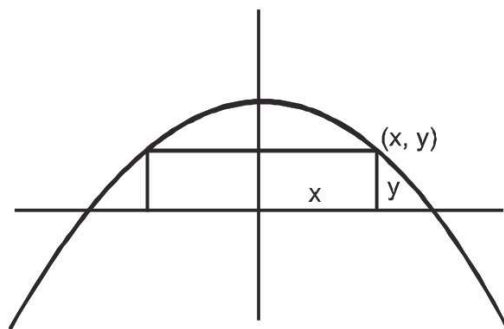
$$h = 10$$

at $l = 10$, cost is minimum

Q72. Find the area of the largest rectangle with lower base on the x-axis & upper vertices on the curve $y = 12 - x^2$.

Ans. 32sq.units

Sol.



Figure

Let A be area of rectangle.

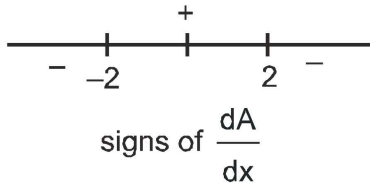
$$A = (2x)(y)$$

$$= 2x(12 - x^2)$$

$$\frac{dA}{dx} = 6(2 + x)(2 - x)$$

At $x = 2$, A has largest value.

$$\text{Largest } A = 32$$



Q73. Given two points A (- 2 , 0) & B (0 , 4) and a line $y = x$. Find the co-ordinates of a point M on this line so that the perimeter of the Δ AMB is least.

Ans. (0 , 0)

Sol. -

Q74. Consider the function $y = f(x) = \ln(1 + \sin x)$ with $-2\pi \leq x \leq 2\pi$. Find

(a) the zeroes of $f(x)$

(b) inflection points if any on the graph

(c) local maxima and minima of $f(x)$

(d) asymptotes of the graph

(e) sketch the graph of $f(x)$ and compute the value of the definite integral $\int_{-\pi/2}^{\pi/2}$.

Ans. (a) $x = -2\pi, -\pi, 0, \pi, 2\pi$, (b) no inflection point, (c) maxima at $x = \pi/2$ and $-3\pi/2$ and no minima,

(d) $x = 3\pi/2$ and $x = -\pi/2$, (e) $-\pi \ln 2$

Sol. -

Q75. Prove that $\sin x + 2x \geq \frac{3x \cdot (x + 1)}{\pi} \forall x \in \left[0, \frac{\pi}{2}\right]$. (Justify the inequality, if any used).

Ans. -

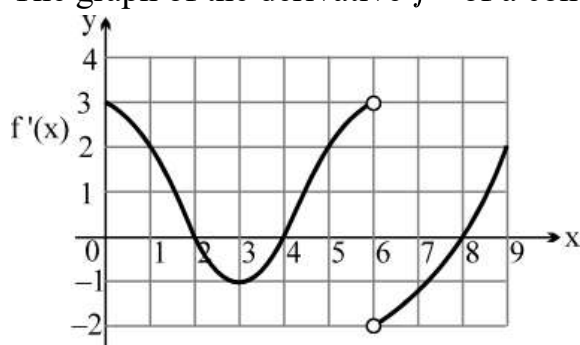
Sol. -

Q76. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is :

Ans. 2

Sol. Let $f(x) = x^4 - 4x^3 + 12x^2 + x - 1$
 $f'(x) = 4x^3 - 12x^2 + 24x$
 $f''(x) = 12x^2 - 24x + 24$
 $= 12(x^2 - 2x + 2) > 0$
 $\Rightarrow f'(x)$ is strictly increasing function
 $\therefore f'(x)$ is cubic polynomial
hence number of roots of $f'(x) = 0$ is 3
 $\Rightarrow$ Number of maximum roots of $f(x) = 0$ are 4
Now $f(0) = -1, f(1) = 9, f(-1) = 15$
 $\Rightarrow f(x)$ has exactly 2 distinct real roots.

Q77. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$. If



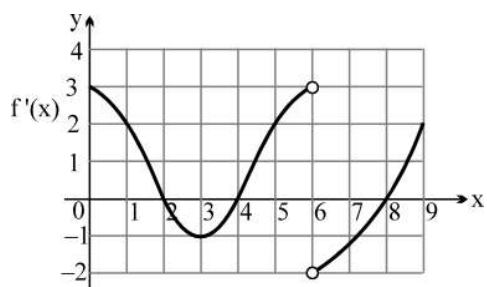
- (i) f is monotonic increasing in the interval $[a, b) \cup (c, d) \cup (e, f]$ and decreasing in $(p, q) \cup (r, s)$.
- (ii) f has a local minima at $x = x_1$ and $x = x_2$.
- (iii) f is concave up in $(l, m) \cup (n, t]$
- (iv) f has inflection point at $x = k$
- (v) number of critical points of $y = f(x)$ is 'w'.

Find the value of $(a + b + c + d + e) + (p + q + r + s) + (l + m + n) + (x_1 + x_2) + (k + w)$.

Ans. 74

Sol. -

Q78. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$

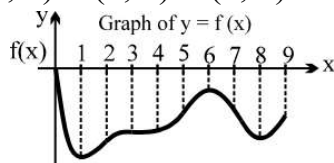


- (i) On what intervals is f increasing or decreasing?

- (ii) At what values of x does f have a local maximum or minimum?
 (iii) On what intervals is f concave upward or downward?
 (iv) State the x -coordinate(s) of the point(s) of inflection.
 (v) Assuming that $f(0) = 0$, sketch a graph of f .

Ans. (i) I in $(1, 6) \cup (8, 9)$ and D in $(0, 1) \cup (6, 8)$; (ii).Min. at $x = 1$ and $x = 8$; L.Max. $x = 6$

(iii) CU in $(0, 2) \cup (3, 5) \cup (7, 9)$ and CD in $(2, 3) \cup (5, 7)$; (iv) $x = 2, 3, 5, 7$



(v) Graph is

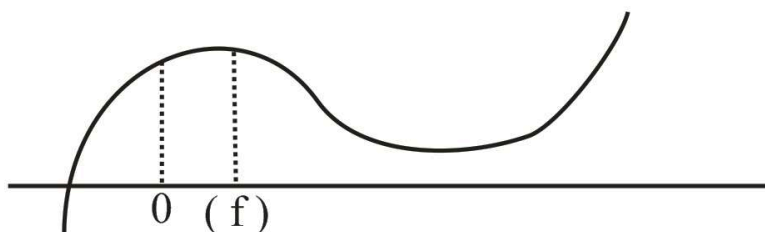
Sol. -

Q79. The value of 'a' for which $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ have a positive point of maximum lies in the interval $(a_1, a_2) \cup (a_3, a_4)$. Find the value of $a_2 + 11a_3 + 70a_4$.

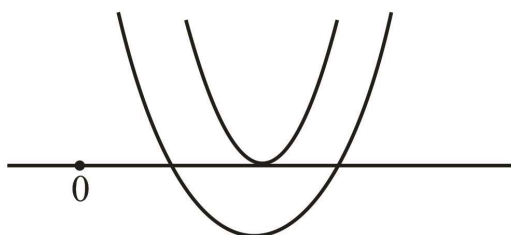
Ans. 320

Sol. $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$

$$f'(x) = 3x^2 + 6(a-7)x + 3(a^2-9)$$



here positive point is of maxima then both roots are greater than '0' of $f'(x)$ then for $f(x)$.



$$(1) d \leq 0$$

$$(2) af(0) > 0$$

$$(3) -\frac{b}{2a} > 0$$

$$(1) 36(a-7)^2 - 4 \times 4^2 \times (a^2-9) \geq 0$$

$$(a^2 - 14a + 49) - a^2 + 9 \geq 0$$

$$a^2 - 14a + 49 - a^2 + 9 \leq 0$$

$$-14a + 58 \geq 0$$

$$14a \leq 58$$

$$a \leq \frac{29}{7}$$

$$(ii) af(0) > 0$$

$$(a^2 - 9) > 0$$

$$a \in (-\infty, -3) \cup (3, \infty)$$

$$(iii) \frac{-6(a-7)}{2 \times 3} > 0$$

$$a < 7$$

by (i), (2) & (3)

Common values of

$$a \in (-\infty, -3) \cup (3, \frac{29}{7})$$

$$a_2 = -3$$

$$a_3 = 3$$

$$a_4 = \frac{29}{7}$$

$$a_2 + 11a_3 + 70a_4$$

$$= -3 + 11(3) + 70\left(\frac{29}{7}\right)$$

$$= -3 + 33 + 290$$

$$= 320$$

Paragraph

Let $f(x) = \frac{1}{1+x^2}$. Let m be the slope, a be the x-intercept and b be the y-intercept of a tangent to $y = f(x)$, then

\n

Q80. Abscissa of the point of contact of the tangent for which m is greatest, is

$$(A) \frac{1}{\sqrt{3}} \quad (B) 1 \quad (C) -1 \quad (D) -\frac{1}{\sqrt{3}}$$

Ans. D

Sol. -

Paragraph

Let $f(x) = \frac{1}{1+x^2}$. Let m be the slope, a be the x-intercept and b be the y-intercept of a tangent to $y = f(x)$, then

\n

Q81. Greatest value of b is

- (A) $\frac{9}{8}$ (B) $\frac{3}{8}$ (C) $\frac{1}{8}$ (D) $\frac{5}{8}$

Ans. A

Sol. -

Paragraph

Let $f(x) = \frac{1}{1+x^2}$. Let m be the slope, a be the x-intercept and b be the y-intercept of a tangent to $y = f(x)$, then

\n

Q82. Abscissa of the point of contact of tangent for which $\frac{1}{2}$ is greatest, is

- (A) $\frac{1}{\sqrt{3}}$ (B) 1 (C) -1 (D) $-\frac{1}{\sqrt{3}}$

Ans. A

Sol. -

Paragraph

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where $a, b \in \mathbb{R}$.

\n

Q83. For $a = 1$ if $y = f(x)$ is strictly increasing $\forall x \in \mathbb{R}$ then maximum range of values of b is

- (A) $\left(-\infty, \frac{1}{3}\right]$ (B) $\left(\frac{1}{3}, \infty\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) $\left(-\infty, \infty\right)$

Ans. C

Sol. -

Paragraph

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where $a, b \in \mathbb{R}$.

\n

Q84. For $b = 1$, if $y = f(x)$ is non monotonic then the sum of all the integral values of $a \in [1, 100]$, is

- (A) 4950 (B) 5049 (C) 5050 (D) 5047

Ans. B

Sol. -

Paragraph

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where $a, b \in \mathbb{R}$.

\n

Q85. If the sum of the base 2 logarithms of the roots of the cubic $f(x) = 0$ is 5 then the value of 'a' is

- (A) -64 (B) -8 (C) -128 (D) -256

Ans. D

Sol. If x_1, x_2 and x_3 are the roots then $\log_2 x_1 + \log_2 x_2 + \log_2 x_3 = 5$

$$\log_2(x_1 x_2 x_3) = 5$$

$$x_1 x_2 x_3 = 32$$

$$-\frac{a}{8} = 32 \Rightarrow a = -256$$

Paragraph

If a continuous function f defined on the real line $\mathbb{R}$, assumes positive and negative values in $\mathbb{R}$ then the equation $f(x) = 0$ has a root in $\mathbb{R}$. For example, if it is known that a continuous function f on $\mathbb{R}$ is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in $\mathbb{R}$.

\nConsider $f(x) = kex - x$ for all real x where k is a real constant.

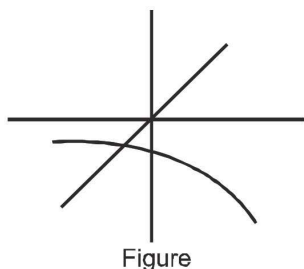
\n

Q86. The line $y = x$ meets $y = ke^x$ for $k \leq 0$ at

- (A) no point (B) one point (C) two points (D) more than two points

Ans. B

Sol.



Paragraph

If a continuous function f defined on the real line $\mathbb{R}$, assumes positive and negative values in $\mathbb{R}$ then the equation $f(x) = 0$ has a root in $\mathbb{R}$. For example, if it is known that a continuous function f on $\mathbb{R}$ is positive at some point and its minimum value is negative then the

equation $f(x) = 0$ has a root in R .

\nConsider $f(x) = kex - x$ for all real x where k is a real constant.

\n

Q87. The positive value of k for which $ke^x - x = 0$ has only one root is

- (A) $1/e$ (B) 1 (C) e (D) $\log_e 2$

Ans. A

Sol. -

Paragraph

If a continuous function f defined on the real line R , assumes positive and negative values in R then the equation $f(x) = 0$ has a root in R . For example, if it is known that a continuous function f on R is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in R .

\nConsider $f(x) = kex - x$ for all real x where k is a real constant.

\n

Q88. For $k > 0$, the set of all values of k for which $kex - x = 0$ has two distinct roots is

- (A) $(0, 1/e)$ (B) $(1/e, 1)$ (C) $(1/e, \infty)$ (D) $(0, 1)$

Ans. A

Sol. -

Paragraph

Consider the function $f(x) = \frac{x^2}{x^2 - 1}$

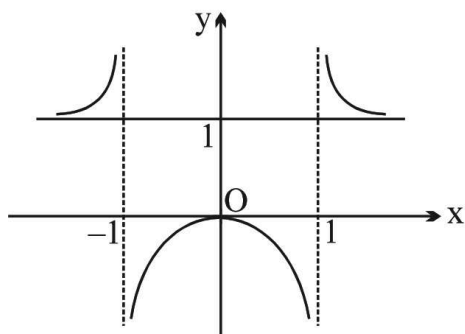
\n

Q89. The interval in which f is increasing is

- (A) $(-1, 1)$ (B) $(-\infty, -1) \cup (-1, 0)$ (C) $(-\infty, \infty) - \{-1, 1\}$ (D) $(0, 1) \cup (1, \infty)$

Ans. B

Sol.



$$y = \frac{x^2}{x^2 - 1}; \text{ not defined at } x = \pm 1$$

$$= 1 + \frac{1}{x^2 - 1}; \quad y' = -\frac{2x}{(x^2 - 1)^2}$$

$$\frac{dy}{dx} = 0 \Rightarrow x = 0 \text{ (point of maxima)}$$

$$\text{as } x \rightarrow 1^+, y \rightarrow \infty; \quad x \rightarrow 1^-, y \rightarrow -\infty$$

$$|||y \text{ as } x \rightarrow -1^+, y \rightarrow -\infty; \quad x \rightarrow -1^-, y \rightarrow \infty$$

The graph of $y = \frac{x^2}{x^2 - 1}$ is as shown verify all alternatives from the graph.

Paragraph

Consider the function $f(x) = \frac{x^2}{x^2 - 1}$

\n

Q90. If f is defined from $\mathbb{R} - \{-1, 1\} \rightarrow \mathbb{R}$ then f is

- (A) injective but not surjective (B) surjective but not injective
(C) injective as well as surjective (D) neither injective nor surjective.

Ans. D

Sol. -

Paragraph

Consider the function $f(x) = \frac{x^2}{x^2 - 1}$

\n

Q91. f has

- (A) local maxima but no local minima (B) local minima but no local maxima
(C) both local maxima and local minima (D) neither local maxima nor local minima.

Ans. A

Sol. -

Paragraph

Consider a rational function $f(x) = \frac{x^2 - 6x + 4}{x^2 + 2x + 4}$ and a quadratic function $g(x) = (1 + m)x^2 - 2(1 + 3m)x - 2(1 + m)$ where m is a parameter.

\n

Q92. Which one of the following statement does not hold good for the rational function $f(x)$?

- (A) It is a continuous function. (B) It has only one asymptote.
(C) It has exactly one maxima and one minima. (D) $f(x)$ is monotonic in $(0, \infty)$

Ans. D

Sol. $f(x) = \frac{x^2 - 6x + 4}{x^2 + 2x + 4} = 1 - \frac{8x}{x^2 + 2x + 4}$

$$f'(x) = -8 \left[\frac{(x^2 + 2x + 4) - x(2x + 2)}{(x^2 + 2x + 4)^2} \right] = -8 \left[\frac{-x^2 + 4}{(x^2 + 2x + 4)^2} \right] = \frac{8(x^2 - 4)}{(x^2 + 2x + 4)^2}$$

$$f(2) = \frac{4 - 12 + 4}{4 + 4 + 4} = \frac{-4}{12} = -\frac{1}{3}$$

Let $y = \frac{8x}{x^2 + 2x + 4}$

$$x^2y + 2(y - 4)x + 4y = 0$$

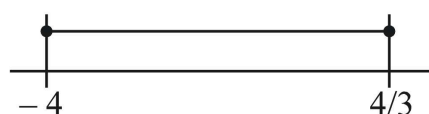
$$x \in \mathbb{R} \Rightarrow D \geq 0$$

$$f(-2) = \frac{4 + 12 + 4}{4 - 4 + 4} = 5$$

$$4(y - 4)^2 - (2y)^2 \geq 0$$

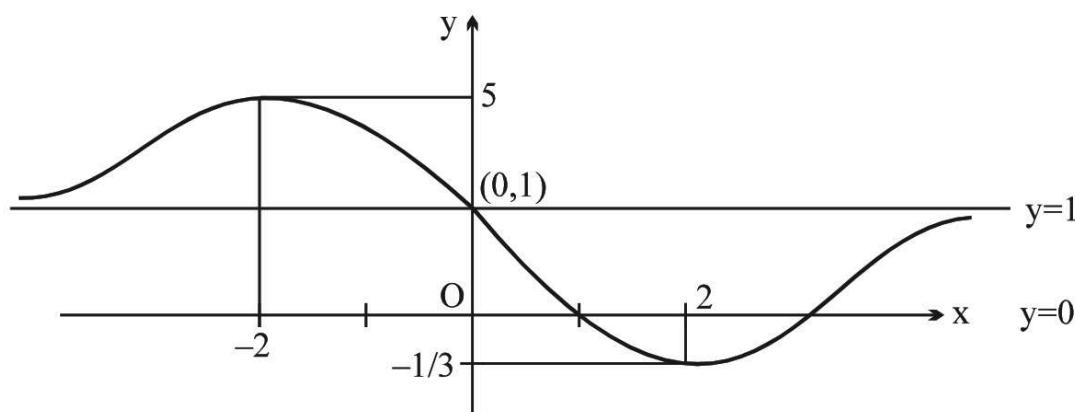
Hence range of $f(x)$ is $\left[-\frac{1}{3}, 5\right]$

$$(3y - 4)(y + 4) \geq 0$$



the graph of $y = f(x)$ is as shown

hence $-\frac{1}{3} \leq f(x) \leq 5$



obviously f is continuous and has $y = 1$ as its asymptote with one maxima and one minima and f is non monotonic in $(0, \infty) \Rightarrow (D)$ is incorrect

Paragraph

Consider a rational function $f(x) = \frac{x^2 - 6x + 4}{x^2 + 2x + 4}$ and a quadratic function $g(x) = (1 + m)x^2 - 2(1 + 3m)x - 2(1 + m)$ where m is a parameter.

\n

Q93. If $\cot(\cot^{-1} f(x)) = k$, has exactly two distinct real solutions then the integral value of 'k' can be

- (A) 0 (B) -1 (C) 1 (D) 5

Ans. A

Sol. -

Paragraph

Consider a rational function $f(x) = \frac{x^2 - 6x + 4}{x^2 + 2x + 4}$ and a quadratic function $g(x) = (1 + m)x^2 - 2(1 + 3m)x - 2(1 + m)$ where m is a parameter.

\n

Q94. If the range of the function $f(x)$ lies between the roots of $g(x) = 0$ then the number of integral values of m equals

- (A) 7 (B) 8 (C) 9 (D) 10

Ans. C

Sol. -

Paragraph

Consider the function $f: (-\infty, \infty) \rightarrow (-\infty, \infty)$ defined by $f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$

\n

Q95. Which of the following is true?

- (A) $(2 + a)^2 f''(1) + (2 - a)^2 f''(-1) = 0$ (B) $(2 - a)^2 f''(1) - (2 + a)^2 f''(-1) = 0$
(C) $f'(1) f'(-1) = (2 - a)^2$ (D) $f'(1) f'(-1) = -(2 + a)^2$

Ans. A

Sol.

$$f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$$

$$f'(x) = \frac{2a(x^2 - 1)}{(x^2 + ax + 1)^2} \text{ and } f'(x) = \frac{4a(-x^3 + 3x + a)}{(x^2 + ax + 1)^3}$$

$$f''(1) = \frac{4a}{(a + 2)^2} \text{ and } f''(-1) = \frac{-4a}{(a - 2)^2}$$

$$\therefore (a + 2)^2 f''(1) + (2 - a)^2 f''(-1) = 0$$

Paragraph

Consider the function $f: (-\infty, \infty) \rightarrow (-\infty, \infty)$ defined by $f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$

\n

Q96. Which of the following is true?

- (A) $f(x)$ is decreasing on $(-1, 1)$ and has a local minimum at $x = 1$
(B) $f(x)$ is increasing on $(-1, 1)$ and has a local maximum at $x = 1$
(C) $f(x)$ is increasing on $(-1, 1)$ but has neither a local maximum and nor a local minimum at $x = 1$

- (D) $f(x)$ is decreasing on $(-1, 1)$ but has neither a local maximum and nor a local minimum at $x = 1$

Ans. A

Sol. As when $x \in (-1, 1)$, $f'(x) < 0$

so $f(x)$ is decreasing on $(-1, 1)$ at $x = 1$

$$f''(1) = \frac{4a}{(a+2)^2} > 0$$

so local minima at $x = 1$.

Paragraph

Consider the function $f: (-\infty, \infty) \rightarrow (-\infty, \infty)$ defined by $f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$

\n

Q97.
$$g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$$

Which of the following is true?

- (A) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$
 (B) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$
 (C) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$
 (D) $g'(x)$ does not change sign on $(-\infty, \infty)$

Ans. B

Sol.
$$g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$$

$$g'(x) = \frac{f'(e^x)}{1+e^{2x}} e^x = \frac{2a(e^{2x} - 1)e^x}{(e^{2x} + ae^x + 1)^2(1+e^{2x})}$$

$$g'(x) > 0 \quad \text{when } x > 0$$

$$g'(x) < 0 \quad \text{when } x < 0$$

Q98. The local maximum value of $x(1-x)^2$, $0 \leq x \leq 2$ is :

- (A) 2 (B) 4/27 (C) 5 (D) 0

Ans. A, B

Sol. Let $f(x) = x(1-x)^2$

$$\begin{aligned} f'(x) &= 1(1-x)^2 + x \times 2(1-x) \times -1 \\ &= (1-x)^2 - 2x(1-x) \end{aligned}$$

$$\begin{aligned}
 f''(x) &= -2(1-x) - 2[1-x+x \times -1] \\
 &= -2 + 2x - 2[1-x-x] \\
 &= -2 + 2x - 2 + 4x \\
 &= 6x - 4
 \end{aligned}$$

For critical points, $F'(x) = 0$

$$\Rightarrow (1-x)^2 - 2x(1-x) = 0$$

$$\Rightarrow (1-x)(1-x-2x) = 0$$

$$\Rightarrow (1-x)(1-3x) = 0$$

$$\Rightarrow (x-1)(3x-1) = 0$$

$$x = 1, \frac{1}{3}$$

$$f'(1) = 6(1) - 4 = 6 - 4 = 2 > 0 \text{ minimum}$$

$$f''\left(\frac{1}{3}\right) = 6 \times \frac{1}{3} - 4 = 2 - 4 = -2 < 0 \text{ minimum}$$

$$f\left(\frac{1}{3}\right) = \frac{1}{3} \left(1 - \frac{1}{3}\right)^2 = \frac{1}{3} \times \frac{4}{9} = \frac{4}{27}$$

Q99.

The function $f(x) = \int_0^x \sqrt{1-t^4} dt$ is such that

(A) it is defined on the interval $[-1, 1]$ (B) it is an increasing function

(C) it is an odd function (D) the point $(0, 0)$ is the point of inflection

Ans. A, B, C, D

Sol. $f'(x) = \sqrt{1-x^4} > 0$ in $(-1, 1) \Rightarrow f$ is

$$\text{Now } f(x) + f(-x) = \int_0^x \sqrt{1-t^4} dt + \int_0^{-x} \sqrt{1-t^4} dt$$

$$\Rightarrow \int_0^x \sqrt{1-t^4} dt + \left(- \int_0^y \sqrt{1-y^4} dy \right)$$

$$= 0 \Rightarrow f(x) \text{ is odd}$$

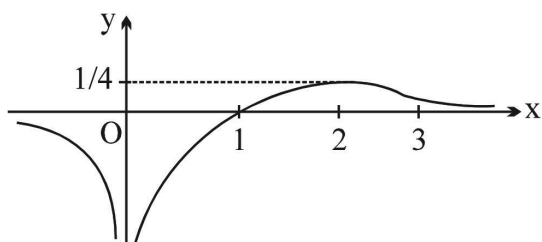
again $f''(x) = \frac{-4x^3}{2\sqrt{1-x^4}}$ which vanishes at $x = 0$ and changes sign $\Rightarrow (0, 0)$ is inflection since f is well defined in $[-1, 1] \Rightarrow A, B, C, D$

Q100. Let $f(x) = \frac{x-1}{x^2}$ then which of the following is correct.

- (A) $f(x)$ has minima but no maxima.
 (B) $f(x)$ increases in the interval $(0, 2)$ and decreases in the interval $(-\infty, 0) \cup (2, \infty)$.
 (C) $f(x)$ is concave down in $(-\infty, 0) \cup (0, 3)$. (D) $x = 3$ is the point of inflection.

Ans. D, C, B

Sol.



$$f'(x) = \frac{2-x}{x^3} \text{ and } f''(x) = \frac{x-3}{x^4}. \text{ Now interpret}$$

Q101. Which of the following functions fail to satisfy the condition of Rolle's theorem on the interval $[-1, 1]$, where $[x]$ denotes the greatest integer less or equal to x and $\{x\}$ denotes the fractional part of x respectively.

- (A) $f(x) = |x| [x]$ (B) $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ (C) $f(x) = \{x\} + \{-x\}$
 (D) $f(x) = |x| - |\sin x|$

Ans. A, C

Sol.

$$(A) \quad f(x) = \begin{cases} 1, & x = 1 \\ 0, & 0 \leq x < 1 \\ x, & -1 \leq x < 0 \end{cases} \Rightarrow \text{not differentiable at } x = 0 \text{ in } (-1, 1)$$

$$(B) \quad f(0) = 0 \text{ and } \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1 \Rightarrow \text{not continuous at } x = 0$$

$$(C) \quad f(x) = \begin{cases} 1, & x \neq I \\ 0, & x \in I \end{cases} \Rightarrow \text{not continuous at } x = 0$$

$$(D) \quad f(x) = \begin{cases} x - \sin x, & 0 \leq x \leq 1 \\ -x + \sin x, & -1 \leq x < 0 \end{cases} \Rightarrow \text{continuous \& differentiable at } x = 0$$

Q102. Which of the following is/are correct?

- (A) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $\tan x = -1$

- (B) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $\tan x = -1$
- (C) Between any two roots of $e^x \cos x = 1$, there exists atleast one root of $e^x \sin x = 1$
- (D) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $e^x \cos x = 1$

Ans. A, B, C

- Sol. (A) Let $f(x) = e^x \cos x - 1$
 $\Rightarrow f'(x) = e^x(\cos x - \sin x) = 0$
 $\Rightarrow \tan x = 1$, between two roots of $f(x)$. (Rolle's theorem)
- (B) Let $g(x) = e^x \sin x - 1$,
 $g'(x) = e^x(\sin x + \cos x) = 0$
 $\Rightarrow \tan x = -1$, between two roots of $g(x)$.
- (C) Let $h(x) = e^{-x} - \cos x$,
 $h'(x) = -e^{-x} + \sin x = 0$
 $\Rightarrow e^{-x} = \sin x$, between two roots of $h(x)$

Q103. A parabola of the form $y = ax^2 + bx + c$ with $a > 0$ intersects the graph of $f(x) = \frac{1}{x^2 - 4}$. Number of possible distinct intersection(s) of these graph is

- (A) 0 (B) 2 (C) 3 (D) 4

Ans. B, C, D

Sol.

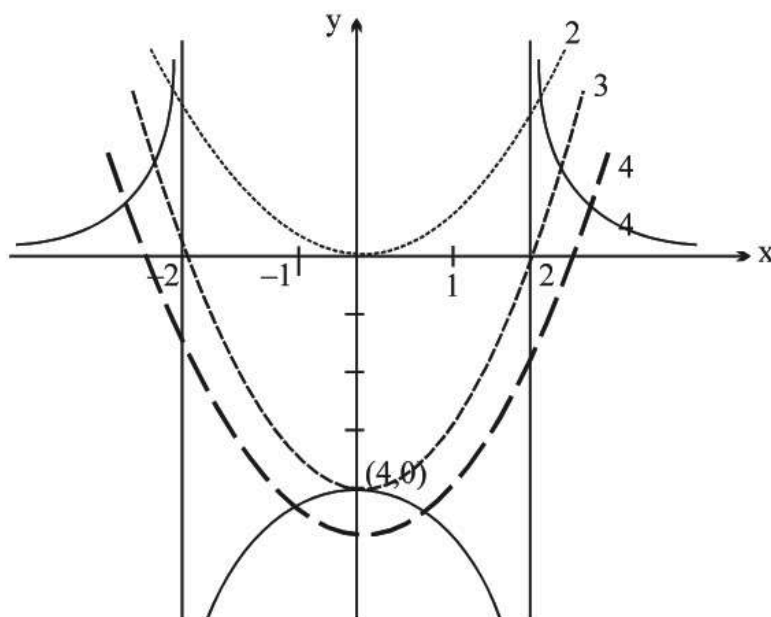


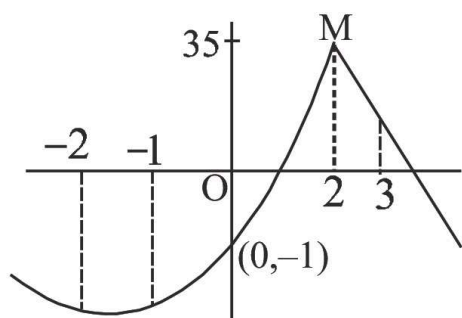
Figure is self explanatory

Q104. If $f(x) = \begin{cases} 3x^2+12x-1 & -1 \leq x \leq 2 \\ 37-x & 2 < x \leq 3 \end{cases}$ then :

- (A) $f(x)$ is increasing on $[-1, 2]$ (B) $f(x)$ is continuous on $[-1, 3]$
 (C) $f'(2)$ does not exist (D) $f(x)$ has the maximum value at $x = 2$

Ans. A, B, C, D

Sol.



Note that $f(x)$ is continuous at $x = 2$

and f is decreasing for $(2, 3)$ and increasing for $[-1, 2]$.

At $x = 2$ f has a maxima]

Q105. Assume that inverse of the function f is denoted by g . Then which of the following statement hold good?

- (A) If f is increasing then g is also increasing.
 (B) If f is decreasing then g is increasing. (C) The function f is injective.
 (D) The function g is onto.

Ans. A, B, C, D

Sol. If f and g are inverse then $(f \circ g)(x) = x$

$$f'[g(x)] g'(x) = 1$$

if f is increasing $\Rightarrow f' > 0 \Rightarrow$ sign of g' is also +ve $\Rightarrow$ (A) is correct

If f is decreasing $\Rightarrow f' < 0 \Rightarrow$ sign of g' is -ve $\Rightarrow$ (B) is false

since f has an inverse $\Rightarrow f$ is bijective $\Rightarrow f$ is injective $\Rightarrow$ (C) is correct

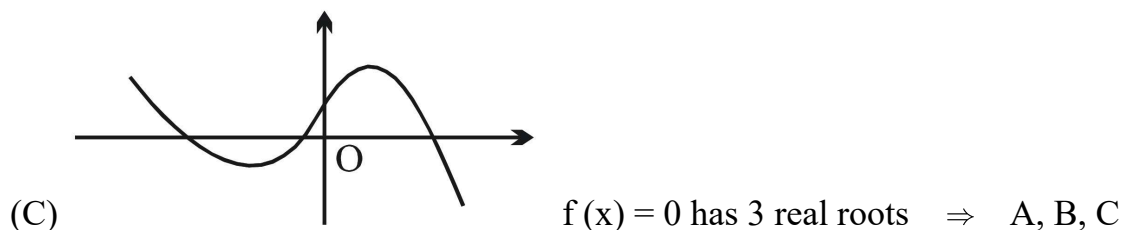
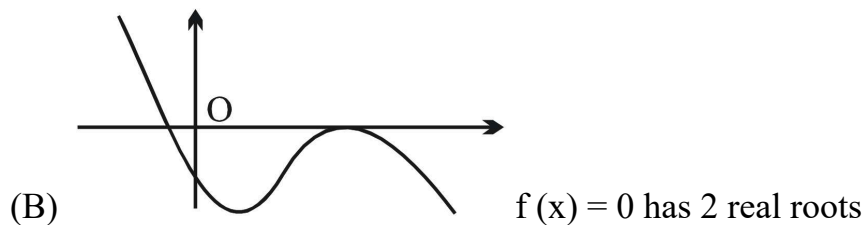
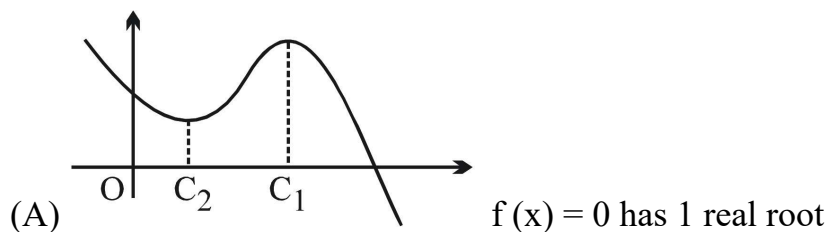
inverse of a bijective mapping is bijective $\Rightarrow g$ is also bijective $\Rightarrow g$ is onto $\Rightarrow$ (D) is correct.

Q106. If $f(x)$ is continuous and derivable $\forall x \in \mathbb{R}$ and $f'(c) = 0$ for exactly 2 real values of ' c ' then the number of real and distinct values of ' d ' for which $f(d) = 0$ can be

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. A, B, C

Sol. If $f'(x) = 0$ has n real roots $\Rightarrow f(x) = 0$ has at most $(n + 1)$ roots

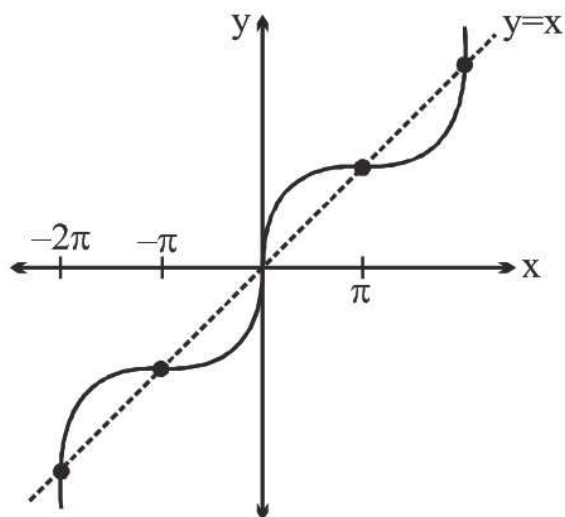


Q107. Consider the function $f : \mathbb{R} \Rightarrow \mathbb{R}$ defined as $f(x) = x + \sin x$. Which of the following is/are the correct statement(s)?

- The function is strictly increasing at every point on $\mathbb{R}$ except at 'x' equal to an odd integral multiple of π where the derivative of $f(x)$ is zero and where the function f is not strictly increasing.
- (A) The function is bounded in every bounded interval but unbounded on whole real line.
- (B) The graph of the function $y = f(x)$ lies in the first and third quadrants only.
- (C) The graph of the function $y = f(x)$ cuts the line $y = x$ at infinitely many points.
- (D)

Ans. B, C, D

Sol.



We have $f'(x) = 1 + \cos x \Rightarrow f$ is strictly increasing and has inflection point at $x = (2n+1)\pi$

Also there is no x for which $f(x)$ is not increasing.

$\Rightarrow$ (A) is not correct

As f is continuous $\forall x \in \mathbb{R}$ hence bounded in every closed interval.

As f is odd hence symmetric w.r.t origin and its graph lies in 1st and 3rd quadrant and $y = x$ cut the graph at infinitely many points

$\Rightarrow$ B, C, D are correct.

Q108.

Ans. -

Sol. -

Q109. Which of the following pair(s) of curves touch each other?

(A) $y^2 = 4x$ and $x^2 + y^2 = 6x - 1$ (B) $xy = 4$ and $x^2 + y^2 = 8$

(C) $y = 2x - 3$ and $y = x^3 - x + 1$ (D) $y = 6 + x - x^2$ and $y = \frac{x+2}{x-2}$

Ans. A, B

Sol. (A) $y^2 = 4x$; $x^2 + y^2 = 6x - 1$

solving, $x^2 + 4x = 6x - 1$

$x^2 - 2x + 1 = 0 \Rightarrow (x-1)^2 = 0 \Rightarrow x = 1$

touch at $(1, 2)$ and $(1, -2)$ B (A) is correct

(B) $xy = 4$; $x^2 + y^2 = 8$

$x^2 + y^2 = 16 \Rightarrow x^2(8 - x^2) = 16$; put $x^2 = t > 0$

$t^2 - 8t + 16 = 0 \Rightarrow (t-4)^2 = 0$

$t = 4 \Rightarrow x^2 = 4 \Rightarrow x = 2$ or $-2 \Rightarrow (2, 2)$ or $(-2, -2)$

(C) Solving the points of intersection are $(1, 1)$; $(-2, -5)$; also $\frac{dy}{dx}$ is same at $(1,$

1).

Hence they touch at (1, 1)

Q110. The co-ordinates of a point on the parabola $2y = x^2$ which is nearest to the point (0, 3) is:

- (A) (2, 2) (B) $(-\sqrt{2}, 1)$ (C) $(\sqrt{2}, 1)$ (D) $(-2, 2)$

Ans. A, D

Sol. Let (a, b) is the point on the parabola from which its distance from (0, 3) is minimum

$$\Rightarrow a^2 = 2b$$

Assume d is the distance

$$d^2 = a^2 + (b - 3)^2 = 2b + (b - 3)^2$$

$$\text{For minimum value of } d \frac{d(d^2)}{db^2} = 0 = 2 + 2(b - 3)$$

$$\Rightarrow b = 2$$

$$\text{and } a^2 = 2b = 4 \quad a = \pm 2$$

Hence require points on the parabola are (2, -2) and (2, 2)

Q111. In the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$, at point (2, -1)

(A) length of subtangent is 7/6. (B) slope of tangent = 6/7

(C) length of tangent = $\sqrt{(85)}/6$ (D) None of these

Ans. A, B, C

Sol. $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$

$$\frac{dx}{dt} = 2t + 3, \quad \frac{dy}{dt} = 4t - 2$$

$$\Rightarrow \frac{dy}{dx} = \left(\frac{dy}{dt} \right) \cdot \left(\frac{dt}{dx} \right) = \left(\frac{4t - 2}{2t + 3} \right)$$

$$\text{The slope of tangent, } \frac{dy}{dx} \text{ at } (2, -1) = \left(\frac{dy}{dt} \right)_{(-2,1)}$$

$$\text{when } x = 2, t^2 + 3t - 8 = 2 \Rightarrow t^2 + 3t - 10 = 0$$

$$\Rightarrow t^2 + 5t - 2t - 10 = 0 \Rightarrow t(t + 5) - 2(t + 5) = 0$$

$$\Rightarrow (t - 2)(t + 5) = 0 \Rightarrow t = 2, \text{ or } -5$$

$$\therefore \frac{dy}{dx} = \left(\frac{4(2) - 2}{2(2) + 3} \right) = \frac{6}{7}$$

When $x = 2$, $y = 1$

$$\therefore -1 = 2t^2 - 2t - 5 \Rightarrow 2t^2 - 2t - 4 = 0$$

$$\Rightarrow 2t^2 - 4t + 2t - 4 = 0$$

$$\Rightarrow 2t(t - 2) + 2(t - 2) = 0 \Rightarrow t = 2, -1$$

So the common value of $t = 2$

$$\therefore \text{Slope of tangent at } (2, -1) = \frac{6}{7}$$

$$\text{Length of tangent} = \sqrt{\left(\frac{y_1}{m}\right)^2 + (y_1)^2}$$

$$\text{Here, } y_1 = -1 \text{ and } m = \frac{6}{7}$$

$$\therefore \text{Length} = \sqrt{\left(\frac{7}{6}\right)^2 + (-1)^2} = \sqrt{\frac{49 + 36}{6}}$$

Q112. The function $f(x) = 2 \ln(x - 2) - x^2 + 4x + 1$ increases in the intervals

(A) $(1, 2)$ (B) $(2, 3)$ (C) $\left[\frac{5}{2}, 3\right]$ (D) $(2, 4)$

Ans. B, C

Sol. -

Q113. If $f(x) = 2x + \cot^{-1} x + \log(\sqrt{1+x^2} - x)$, then $f(x)$:

(A) increases in $[0, \infty)$ (B) decreases in $[0, \infty)$

(C) neither increases nor decreases in $[0, \infty)$ (D) increases in $(-\infty, \infty)$

Ans. A, D

Sol. -

Q114. Let $g(x) = 2f(x/2) + f(1 - x)$ and $f'(x) < 0$ in $0 \leq x \leq 1$ then $g(x)$:

(A) decreases in $\left[0, \frac{2}{3}\right)$ (B) decreases $\left(\frac{2}{3}, 1\right]$ (C) increases in $\left[0, \frac{2}{3}\right)$

(D) increases in $\left(\frac{2}{3}, 1\right]$

Ans. C, B

Sol. -

Q115. If $f(x) = \tan^{-1} x - (1/2) \ln x$. Then

(A) the greatest value of $f(x)$ on $[1/\sqrt{3}, \sqrt{3}]$ is $\pi/6 + (1/4) \ln 3$

(B) the least value of $f(x)$ on $[1/\sqrt{3}, \sqrt{3}]$ is $\pi/3 - (1/4) \ln 3$

(C) $f(x)$ decreases on $(0, \infty)$ (D) $f(x)$ increases on $(-\infty, 0)$

Ans. C, B, A

Sol. -

Q116. If $f(x) = \log(x-2) - \frac{1}{x}$, then

- (A) $f(x)$ is M.I. for $x \in (2, \infty)$ (B) $f(x)$ is M.I. for $x \in [-1, 2]$
 (C) $f(x)$ is always concave downwards (D) $f^{-1}(x)$ is M.I. wherever defined

Ans. A, C, D

Sol. $f(x) = \ln(x-2) - \frac{1}{x}$

$$f'(x) = \frac{1}{x-2} + \frac{1}{x^2} = \frac{x^2 + x - 2}{x^2(x-2)} = \frac{x^2 + 2x - x - 2}{x^2(x-2)}$$

$$= \frac{(x+2)(x-1)}{x^2(x-2)}.$$

As $\ln(x-2)$ is defined when $x > 2$

$\Rightarrow f(x)$ is M.I. for $x \in (2, \infty)$

$\Rightarrow f^{-1}(x)$ is M.I. wherever defined

Also $f''(x) = \frac{-1}{(x-2)^2} - \frac{2}{x^3} < 0$

$\Rightarrow f(x)$ is always concave downward

Q117. The function $f(x) = x^{1/3}(x-1)$

- (A) has 2 inflection points.
 (B) is strictly increasing for $x > 1/4$ and strictly decreasing for $x < 1/4$.
 (C) is concave down in $(-1/2, 0)$.
 (D) Area enclosed by the curve lying in the fourth quadrant is .

Ans. D, C, B, A

Sol. -

Q118. $f''(x) > 0$ for all $x \in [-3, 4]$, then which of the following are always true?

- (A) $f(x)$ has a relative minimum on $(-3, 4)$ (B) $f(x)$ has a minimum on $[-3, 4]$
 (C) $f(x)$ is concave upwards on $[-3, 4]$
 (D) if $f(3) = f(4)$ then $f(x)$ has a critical point on $[-3, 4]$

Ans. C, B, D

Sol. (A) $f(x)$ has no relative minimum on $(-3, 4)$

(B) $f(x)$ is continuous function on $[-3, 4]$

$\Rightarrow f(x)$ has min. and max. on $[-3, 4]$ by IVT

(C) $f''(x) > 0$

$\Rightarrow f(x)$ is concave upwards on $[-3, 4]$

(D) $f(3) = f(4)$

By Rolle's theorem

$$\exists c \in (3, 4), \text{ where } f'(c) = 0$$

$$\Rightarrow \exists \text{ critical point on } [-3, 4]$$

Q119. The function $f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x < 1 \\ (x-2)^2 & \text{if } x \geq 1 \end{cases}$

(A) is continuous for all $x \in \mathbb{R}$ (B) is continuous but not differentiable $\forall x \in \mathbb{R}$

(C) is such that $f'(x)$ change its sign exactly twice

(D) has two local maxima and two local minima.

Ans. A, B, D

Sol. -

Q120. Consider the function $f(x) = [\cos(\tan^{-1}(\sin(\cot^{-1}x)))]^2$. Which of the following is correct?

(A) range of f is $(0, 1)$ (B) f is even (C) $f'(0) = 0$

(D) the line $y = 1$ is asymptotes to the graph $y = f(x)$

Ans. B, C, D

Sol. -

Q121. The function $f(x) = \int_{-1}^x t(e^t - 1)(t-1)(t-2)^3(t-3)^5 dt$ has a local minimum at $x =$

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B, D

Sol. -

Q122. Consider a real valued continuous function $f(x)$ defined on the interval $[a, b]$. Which of the following statements does not hold(s) good?

(A) If $f(x) \geq 0$ on $[a, b]$ then $\int_a^b f(x)dx \leq \int_a^b f^2(x)dx$.

(B) If $f(x)$ is increasing on $[a, b]$, then $f^2(x)$ is increasing on $[a, b]$.

(C) If $f(x)$ is increasing on $[a, b]$, then $f(x) \geq 0$ on (a, b) .

(D) If $f(x)$ attains a minimum at $x = c$ where $a < c < b$, then $f'(c) = 0$.

Ans. A, B, C, D

Sol. -

Q123. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$ (a is a finite quantity), where $[\cdot]$ denotes greatest integer function and $f(x)$ is a non constant continuous function, then

- (A) $\lim_{x \rightarrow a} f(x)$ is an integer. (B) $\lim_{x \rightarrow a} f(x)$ need not be an integer.
 (C) $f(x)$ has a local minimum at $x = a$ (D) $f(x)$ has a local maximum at $x = a$.

Ans. A, C

Sol. -

Q124. Let the function $g : (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2 \tan^{-1}(eu) - \frac{\pi}{2}$. Then, g is

- (A) even and is strictly increasing in $(0, \infty)$
 (B) odd and is strictly decreasing in $(-\infty, \infty)$
 (C) odd and is strictly increasing in $(-\infty, \infty)$
 (D) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

Ans. C

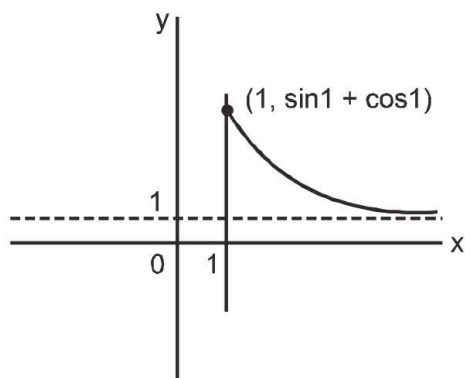
Sol. -

Q125. For the function $f(x) = x \cos \frac{1}{x}, x \geq 1$,

- (A) for at least one x in the interval $[1, \infty)$, $f(x+2) - f(x) < 2$ (B) $\lim_{x \rightarrow \infty} f'(x) = 1$
 (C) for all x in the interval $[1, \infty)$, $f(x+2) - f(x) > 2$
 (D) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$

Ans. B, C, D

Sol.



$$f(x) = x \cos \frac{1}{x}, x \geq 1$$

$$\Rightarrow f'(x) = \frac{1}{x} \sin \frac{1}{x} + \cos \frac{1}{x}$$

$$\Rightarrow f''(x) = -\frac{1}{x^3} \cos\left(\frac{1}{x}\right)$$

Now $\lim_{x \rightarrow \infty} f'(x) = 0 + 1 = 1 \Rightarrow$ option 'B' is correct

$$x \in [1, \infty) \Rightarrow \frac{1}{x} \in (0, 1]$$

$\Rightarrow f'(x) < 0 \Rightarrow$ option 'D' is correct

As $f'(1) = \sin 1 + \cos 1 > 1$

$f'(x)$ is strictly decreasing and $\lim_{x \rightarrow \infty} f'(x) = 1$

so graph of $f(x)$ is as below

Now in $[x, x+2]$, $x \in [1, \infty)$, $f(x)$ is continuous and differentiable

$$\text{so by LMVT, } f'(x) = \frac{f(x+2) - f(x)}{2}$$

as $f'(x) > 1$ for all $x \in [1, \infty)$

$$\Rightarrow \frac{f(x+2) - f(x)}{2} > 1 \Rightarrow f(x+2) - f(x) > 2$$

for all $x \in [1, \infty)$

Q126. Match the following column :

	Column – I		Column – II
(A)	$f(x) = \frac{\sin x}{e^x}, x \in [0, \pi]$	(P)	Condition of Rolle's theorem are satisfied
(B)	$f(x) = \operatorname{sgn}((e^x - 1) \ln x), x \in \left[\frac{1}{2}, \frac{1}{2}\right]$	(Q)	Condition of LMVT are satisfied
(C)	$f(x) = (x-1)^{2/5}, x \in [0, 3]$	(R)	At least one condition of Rolle's theorem is not satisfied.
(D)	$f(x) = \begin{cases} x \left(\frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} \right), & x \in [-1, 1] - \{0\} \\ 0, & x = 0 \end{cases}$	(S)	At least one condition

		of LMVT is not satisfied
--	--	--------------------------------

- (A) (A) P,S (B) R,Q (C) R,S (D) R, S (B) (A) P,Q (B) R,S (C) R,S (D) R, S
 (C) (A) P,Q (B) R,S (C) R,Q (D) P, S (D) (A) P,Q (B) P,S (C) Q,S (D) R, S

Ans. B

Sol. (A) $f(x)$ is continuous and differentiable $f(0) = f(p)$

Hence condition in Rolle's theorem and LMVT are satisfied.

(B) $f(1^-) = -1$, $f(1) = 0$, $f(1^+) = 1$

$f(x)$ is not continuous at $x = 1$, belonging to $\left[\frac{1}{2}, \frac{3}{2}\right]$

Hence, atleast one condition in LMVT and Rolle's theorem is not satisfied

(C) $f'(x) = 2/5 (x-1)^{-3/5}$, $x \neq 1$

At $x = 1$, $f(x)$ is not differentiable.

Hence at least one condition in LMVT and Rolle's theorem is not satisfied.

(D) At $x = 0$

$$L.H.D. = \lim_{x \rightarrow 0^-} \frac{x \left(\frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} \right) - 0}{x - 0} = \frac{0 - 1}{0 + 1} = -1$$

R.H.D. = 1

At $x = 0$, $f(x)$ is not differentiable

Hence at least one condition in LMVT and Rolle's theorem is not satisfied.

Q127. In the following $[x]$ denotes the greatest integer less than or equal to x .

Match the functions in Column I with the properties in Column II.

	Column I		Column II
(A)	$x x $	(P)	continuous in $(-1, 1)$
(B)	$\sqrt{ x }$	(Q)	differentiable in $(-1, 1)$
(C)	$x + [x]$	(R)	strictly increasing in $(-1, 1)$
(D)	$ x - 1 + x + 1 $	(S)	non differentiable at least at one point in $(-1, 1)$

(A) (A) P,Q,R (B) P,S,R (C) Q, R,S (D) P, Q (B) (A) P,Q,R (B) P,S (C) S (D) P, Q,R

(C) (A) P,Q,S (B)S (C) R,Q (D) P, Q (D) (A) P,Q,R (B) P,S (C) R,S (D) P, Q

Ans. D

Sol. -